

THE TATE CONJECTURE FOR POWERS OF ABELIAN FOURFOLDS AND THE HODGE CONJECTURE FOR POWERS OF CM FOURFOLDS

NINGYI LI

ABSTRACT. We prove the Tate conjecture in every codimension on every power of an abelian variety of dimension at most four over a finite field. We also prove the Hodge conjecture for every power of a complex CM abelian fourfold. The finite field proof uses Broe’s theorem and the classification of Frobenius relations for geometrically simple fourfolds. For dihedral pairs of abelian surfaces, we construct the required cycles from families of abelian threefolds, Gross–Schoen heights, and polarization contractions. The Hodge proof uses Markman’s theorem on Weil classes and Milne’s criterion. Over finite fields, the Tate theorem gives standard conjecture D and the expected zeta function pole orders on every power.

CONTENTS

1. Introduction	1
2. Notation and conventions	3
3. Hodge classes on powers of CM fourfolds	4
4. Broe’s theorems	5
5. Frobenius tori and the stabilizer criterion	6
6. Geometrically simple fourfolds	10
7. Mixed isogeny types	11
8. Dihedral pairs of abelian surfaces	13
9. Curves generating the third cohomology	15
10. The mixed dihedral class	21
11. Applications	25
Acknowledgements	27
Appendix A. Dihedral divisor modules	27
References	28

1. INTRODUCTION

We prove the Tate conjecture for all powers of abelian varieties of dimension at most four over finite fields, and the Hodge conjecture for all powers of complex CM abelian fourfolds.

Tate conjecture. Let $\mathbf{F}_q$ have characteristic p , and let F_q denote geometric Frobenius. The Tate conjecture asserts that Frobenius invariant classes in even ℓ -adic cohomology, with the corresponding Tate twist, are spanned by algebraic cycles. Tate proves this for divisors [Tat66, Theorem 4, pp. 143–144], and Broe proves it in codimension two for abelian fourfolds [Bro26, Theorem 1.1]. Our first theorem treats all powers.

2020 *Mathematics Subject Classification.* Primary 14C25; Secondary 14K15, 14F20, 11G10.

Key words and phrases. Tate conjecture, Hodge conjecture, abelian fourfolds, finite fields, CM abelian varieties, algebraic cycles, Frobenius tori, Mumford–Tate groups, Weil classes.

Theorem 1.1. *Let $A/\mathbf{F}_q$ be an abelian variety of dimension at most four. For every $n \geq 1$, every $0 \leq r \leq n \dim A$, and every prime $\ell \neq p$, the cycle class map*

$$\mathrm{CH}^r(A^n)_{\mathbf{Q}} \otimes_{\mathbf{Q}} \mathbf{Q}_{\ell} \longrightarrow H_{\mathrm{\acute{e}t}}^{2r}(A_{\overline{\mathbf{F}}_q}^n, \mathbf{Q}_{\ell}(r))^{F_q=1}$$

is surjective.

Passage to powers. For $n = 1$, Broe's theorem, Tate's divisor theorem, and hard Lefschetz give the conclusion of Theorem 1.1; see Theorem 4.1. On a power of A , a Frobenius eigenvector can occur on several Künneth factors. The resulting Tate classes are governed by integral relations among normalized Frobenius eigenvalues. The stabilizer criterion in Proposition 5.4 reduces Theorem 1.1 to constructing algebraic tensors whose weights generate the even relation lattice.

A dihedral pair $A = S \times T$ from Lemma 8.1 illustrates the passage to the square. After a finite constant extension, choose normalized Frobenius eigenvalues $x_1^{\pm 1}, x_2^{\pm 1}$ on S and $y_1^{\pm 1}, y_2^{\pm 1}$ on T such that

$$y_1 = x_1 x_2, \quad y_2 = x_1 x_2^{-1},$$

where x_1, x_2 are multiplicatively independent; see (10.8). The full relation lattice has basis

$$s_1 = (1, 1, -1, 0), \quad s_2 = (1, -1, 0, -1).$$

Its even part consists of $(a+b, a-b, -a, -b)$ with $a \equiv b \pmod{2}$. Every weight in the exterior algebra of $H^1(A)$ has coordinates in $\{-1, 0, 1\}$. An even relation in this range satisfies $a + b = a - b = 0$, so the only relation weight in the even cohomology of A is zero. On A^2 , the two generators

$$r_1 = s_1 + s_2 = (2, 0, -1, -1), \quad r_2 = s_1 - s_2 = (0, 2, -1, 1)$$

occur as weights in $H^4(A^2)(2)$. Proposition 10.4 constructs algebraic cycles with these weight components, and Proposition 10.5 proves that they generate the even relation lattice integrally.

Correspondences from curves. The construction of the dihedral cycles uses the following theorem.

Theorem 1.2. *Let $Y/\overline{\mathbf{F}}_p$ be an abelian threefold with Newton slopes*

$$0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1.$$

There are a finite disjoint union T of smooth projective curves and a correspondence $\Gamma \in \mathrm{CH}^2(T \times Y)_{\mathbf{Q}}$, both defined over a finite field, such that, for every $\ell \neq p$,

$$\Gamma_* : H^1(T, \mathbf{Q}_{\ell})(-1) \longrightarrow H^3(Y, \mathbf{Q}_{\ell})$$

is surjective.

Idea of the proof. We first prove Theorem 1.2. At the geometric generic point of the Newton stratum, the Gross–Schoen height calculation gives a correspondence with nonzero primitive component. Monodromy makes its image the full primitive cohomology. Spreading the correspondence and using density of isogeny classes gives a single collection of curves and cycles for every $\ell \neq p$ in each prescribed isogeny class; Section 9.3 gives the construction.

For a simple fourfold with commutative endomorphism algebra and nonzero relation lattice, the classification of Arango-Piñeros, Frengley, and Vemulapalli gives a primitive generator realized on the fourfold [APFV25, Appendix A.1, Lemma A.1 and Tables A.3–A.6]; Lemma 6.1 gives the precise reduction. Broe's theorem supplies its algebraic class. The mixed isogeny types reduce to products involving elliptic curves and to the dihedral pair $S \times T$, with S almost ordinary and T ordinary. For a supersingular elliptic curve E , applying Theorem 1.2 to $S \times E$ and then using Tate's homomorphism theorem gives the residual cycles on $S \times T \times E$ in Proposition 10.2. Contracting two such cycles along the S - and E -factors produces the weight components in Proposition 10.4.

Hodge conjecture. For a smooth projective complex variety X , put

$$\mathrm{Hdg}^r(X) = H^{2r}(X, \mathbf{Q}(r))^{(0,0)}.$$

Theorem 1.3. *Let $A/\mathbf{C}$ be a CM abelian fourfold. For every $n \geq 1$ and every $0 \leq r \leq 4n$, the cycle class map*

$$\mathrm{CH}^r(A^n)_{\mathbf{Q}} \longrightarrow \mathrm{Hdg}^r(A^n)$$

is surjective.

The proof of Theorem 1.3 uses Weil classes and the classification of Hodge groups. It is independent of the finite field construction. Both proofs identify the relevant group as the stabilizer of algebraic tensors.

For a simple CM fourfold, Orr’s rank bound [Orr15, Theorem 1.1] shows that its Mumford–Tate group either equals its Lefschetz group or has codimension one in it. When the codimension is one, Milne’s classification identifies the quotient with the determinant character of a Weil structure [Mil22, Lemma 1.9]. Markman’s algebraicity theorem for the Weil plane [Mar25, Corollary 1.6.1 and its proof] and Milne’s criterion [Mil22, Theorem 1.8] then give the Hodge conjecture on all powers. For nonsimple CM fourfolds, the theorem follows from Moonen and Zarhin’s classification [MZ99, Theorem 0.1(iv)] and Milne’s theorem for products of a threefold and a CM elliptic curve [Mil22, Corollary 1.11].

Related results. The Tate conjecture for arbitrary products of elliptic curves is due to Spieß; see [Mil22, Theorem 4.1(c)]. Zarhin classifies the exceptional simple threefolds [Zar15, Theorem 1.1 and Corollary 4.6], and Milne treats all products of a neat simple threefold and an elliptic curve [Mil22, Theorem 4.2]. We use Spieß’s, Zarhin’s, and Milne’s theorems for the products involving elliptic curves in Section 7.

In characteristic zero, Markman’s Corollary 1.6.1 proves the Hodge conjecture for every abelian fourfold [Mar25, Corollary 1.6.1]. Floccari and Fu prove it on all powers of every fourfold of Weil type with discriminant one [FF26, Theorem 1.2]. Theorem 1.3 treats all CM fourfolds, including those of Weil type with arbitrary discriminant.

Consequences. For every power of an abelian variety of dimension at most four, Corollary 11.2 gives equality of numerical and homological equivalence, independence of cycle class kernels from ℓ , and the zeta function pole formula. Proposition 11.1 gives the cycle class statement for motivic summands. Broe’s theorem on standard conjecture D [Bro26, Corollary 1.2] also yields the applications to ample divisors and products of surfaces in Corollary 11.3 and Theorem 11.5.

Overview of the contents. Section 2 fixes the motivic conventions and records descent over finite fields and coefficient extensions. Section 3 proves Theorem 1.3, and Section 4 records the forms of Broe’s theorems used here. Section 5 develops the Frobenius relation lattice and the stabilizer criterion. Sections 6 and 7 treat the simple and mixed isogeny types, and Lemma 8.1 classifies the dihedral pair. Section 9 proves Theorem 1.2. Section 10 constructs the dihedral cycles and completes the proof of Theorem 1.1. Section 11 separates the consequences for all powers from those of Broe’s theorem on standard conjecture D. Appendix A supplies the divisor calculation used in Lemma 8.1.

2. NOTATION AND CONVENTIONS

We use contravariant Chow motives and rational coefficients for Chow groups and motives. If X and Y are smooth projective, with $\dim X = d$, a cycle $\Gamma \in \mathrm{CH}^c(X \times Y)_{\mathbf{Q}}$ acts covariantly on cohomology by

$$\Gamma_* : H^i(X, \mathbf{Q}_{\ell}(j)) \longrightarrow H^{i+2(c-d)}(Y, \mathbf{Q}_{\ell}(j + c - d)). \quad (2.1)$$

It sends $\mathrm{CH}^r(X)_{\mathbf{Q}}$ to $\mathrm{CH}^{r+c-d}(Y)_{\mathbf{Q}}$. Numerically trivial morphisms form a two-sided tensor ideal, so Chow correspondences preserve numerical triviality.

For a Chow motive M , write $R_\ell(M)$ for its ℓ -adic realization. Write $\mathbf{L} = \mathbf{Q}(-1)$ for the Lefschetz motive. Isogenies of abelian varieties induce isomorphisms of rational Chow motives.

Let $X/\mathbf{F}_q$ be smooth and projective, and let F_q denote geometric Frobenius. A class in $H^{2r}(X_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(r))$ is a *Tate class* if it is fixed by some power of F_q . We write

$$\mathcal{T}_\ell^r(X_{\overline{\mathbf{F}}_q}) = \bigcup_{n \geq 1} H^{2r}(X_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(r))^{F_q^n = 1}$$

and let $\mathcal{A}_\ell^r(X_{\overline{\mathbf{F}}_q})$ be the image of $\mathrm{CH}^r(X_{\overline{\mathbf{F}}_q})_{\mathbf{Q}} \otimes_{\mathbf{Q}} \mathbf{Q}_\ell$. Every algebraic class is Tate. The Tate conjecture asserts $\mathcal{T}_\ell^r = \mathcal{A}_\ell^r$. Lemma 2.1 identifies this with the invariant formulation over every finite extension.

For a smooth projective X/k , write $D_\ell^r(X)$ for the assertion that every numerically trivial element of $\mathrm{CH}^r(X)_{\mathbf{Q}}$ has zero class in $H^{2r}(X, \mathbf{Q}_\ell(r))$. Homological triviality implies numerical triviality, so $D_\ell(X)$ is the conjunction of $D_\ell^r(X)$ over r .

For abelian varieties, we use the Chow–Künneth decomposition and the identification of h^i with the i th categorical symmetric power of h^1 [Anc21, Theorem 4.1(1),(2), p. 14]. The categorical symmetry on two factors of h^1 realizes minus the ordinary switch. Inverse Lefschetz operators are algebraic [Anc21, Theorem 4.1(4), p. 14].

For $N = h^1(B)$, write

$$\mathrm{Sym}_{\mathrm{gr}}^2 N = \left(N^{\otimes 2}, \frac{1 - \tau_N}{2} \right). \quad (2.2)$$

Here τ_N is the categorical switch. Its realization is minus the ordinary switch, so $\mathrm{Sym}_{\mathrm{gr}}^2 N$ realizes the ordinary symmetric square. We write $a \odot b = (a \otimes b + b \otimes a)/2$.

2.1. Descent.

Lemma 2.1. *Let $X/\mathbf{F}_q$ be smooth and projective. Let $t \in H^{2r}(X_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(r))$ satisfy $F_q t = t$. If $z_n \in \mathrm{CH}^r(X_{\mathbf{F}_{q^n}})_{\mathbf{Q}} \otimes_{\mathbf{Q}} \mathbf{Q}_\ell$ has class t and $\pi_n : X_{\mathbf{F}_{q^n}} \rightarrow X$ is the projection, then*

$$z = \frac{1}{n} (\pi_n)_* z_n \quad (2.3)$$

has class t over $\mathbf{F}_q$.

Proof. After geometric base change, the class of $(\pi_n)_* z_n$ is $\sum_{j=0}^{n-1} F_q^j(t) = nt$. $\square$

Lemma 2.1 also applies with base field $\mathbf{F}_{q^m}$ and Frobenius F_q^m .

For descent of coefficients, let $X/\mathbf{F}_q$ be smooth and projective, and let $\Omega/\mathbf{Q}_\ell$ be a finite extension. For every $N \geq 1$, the inclusion

$$\ker(F_q^N - 1) \otimes_{\mathbf{Q}_\ell} \Omega \subset \mathcal{A}_\ell^r(X_{\overline{\mathbf{F}}_q}) \otimes_{\mathbf{Q}_\ell} \Omega$$

implies $\ker(F_q^N - 1) \subset \mathcal{A}_\ell^r(X_{\overline{\mathbf{F}}_q})$ by faithful flatness. Here the kernel is taken in $H^{2r}(X_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(r))$.

3. HODGE CLASSES ON POWERS OF CM FOURFOLDS

For a complex abelian variety B , let $L(B)$ be its Lefschetz group, $\mathrm{MT}(B)$ its Mumford–Tate group, and $G_{\mathrm{alg}}(B)$ the subgroup of $L(B)$ fixing all algebraic classes on powers of B . These groups satisfy

$$\mathrm{MT}(B) \subset G_{\mathrm{alg}}(B) \subset L(B).$$

The Hodge conjecture for all powers of B is equivalent to

$$G_{\mathrm{alg}}(B) = \mathrm{MT}(B); \quad (3.1)$$

see [Mil22, Section 1(b), p. 3, and Appendix A(a), p. 20]. Write $S(B)$ and $\text{Hg}(B)$ for the multiplier kernels of $L(B)$ and $\text{MT}(B)$, respectively. We call B *neat* if $\text{MT}(B) = L(B)$.

Proof of Theorem 1.3. The assertion is invariant under isogeny.

Simple fourfolds. Suppose that A is simple. Then $E = \text{End}^0(A)$ is a CM field of degree eight. For a CM field F , write U_F for the torus defined by $u\bar{u} = 1$. We have

$$S(A) = U_E, \quad L(A) = w(\mathbf{G}_m)U_E, \quad \dim S(A) = 4, \quad \dim L(A) = 5,$$

where $w(\mathbf{G}_m)$ is the scalar subgroup. If A is neat, divisor classes generate the Hodge rings of all powers [Mil22, Proposition 1.1].

Suppose that $\text{MT}(A) \subsetneq L(A)$. Orr's bound [Orr15, Theorem 1.1] gives

$$\dim \text{MT}(A) \geq 2 + \log_2(\dim A) = 4.$$

Both groups are connected tori. Hence $\dim \text{MT}(A) = 4$ and $\dim \text{Hg}(A) = 3$. By [Mil22, Lemma 1.9], there is a unique imaginary quadratic subfield $K \subset E$ such that

$$\text{Hg}(A) = \ker(\text{Nm}_{E/K} : U_E \longrightarrow U_K). \quad (3.2)$$

The determinant representation on

$$W_K(A) = \left(\bigwedge_K^4 H^1(A, \mathbf{Q}) \right) (2) \subset H^4(A, \mathbf{Q}(2)) \quad (3.3)$$

is induced by $\text{Nm}_{E/K}$ on U_E and is trivial on scalars. Thus $W_K(A)$ consists of Hodge classes, and (A, K) is of Weil type with signature $(2, 2)$ [Mil22, Proposition 1.3]. Its determinant character gives an exact sequence

$$1 \longrightarrow \text{MT}(A) \longrightarrow L(A) \xrightarrow{\rho_K} U_K \longrightarrow 1. \quad (3.4)$$

Markman's theorem gives algebraicity of every class in $W_K(A)$ [Mar25, Corollary 1.6.1 and its proof]. Therefore

$$\text{MT}(A) \subset G_{\text{alg}}(A) \subset \ker \rho_K = \text{MT}(A).$$

This proves (3.1); see also [Mil22, Theorem 1.8].

Products. Suppose that A is nonsimple. If every simple isogeny factor has dimension at most two, divisors generate the Hodge ring of A^n for every $n \geq 1$ by Moonen and Zarhin's Theorem 0.1(iv) [MZ99, Theorem 0.1(iv) and Introduction, cases (a)–(d)]. Otherwise $A \sim B \times E_0$, where B is a simple threefold and E_0 is a CM elliptic curve. The Hodge conjecture holds for every $B^r \times E_0^s$ by Milne's Corollary 1.11 [Mil22, Corollary 1.11]. Take $r = s = n$. $\square$

Corollary 3.1. *Let $A/\mathbf{C}$ be a CM abelian fourfold. Every $\text{MT}(A)$ -invariant rational tensor in the rigid tensor category generated by $H^1(A, \mathbf{Q})$ and $\mathbf{Q}(1)$ is induced by an algebraic correspondence between powers of A .*

Proof. Polarizations, duality, and the Chow–Künneth projectors identify the tensor with a Hodge class on a power of A . Apply Theorem 1.3. $\square$

4. BROE'S THEOREMS

We record the consequences of Broe's theorems used in the proof.

Theorem 4.1 (Broe [Bro26, Theorem 1.1]). *Let $B/\mathbf{F}_q$ be an abelian variety of dimension at most four. For every r and every prime $\ell \neq p$, the cycle class map*

$$\text{CH}^r(B)_{\mathbf{Q}} \otimes_{\mathbf{Q}} \mathbf{Q}_{\ell} \longrightarrow H_{\text{ét}}^{2r}(B_{\overline{\mathbf{F}}_q}, \mathbf{Q}_{\ell}(r))^{F_q=1}$$

is surjective.

Reduction of Theorem 4.1 to Broe's theorem. When $\dim B = 4$ and $r = 2$, the assertion is [Bro26, Theorem 1.1]. For an abelian variety $B/\mathbf{F}_q$ of dimension $d \geq 2$, choose an ample class ϑ defined over $\mathbf{F}_q$. Hard Lefschetz gives a Frobenius-equivariant isomorphism

$$L_{\vartheta}^{d-2} : H_{\text{ét}}^2(B_{\overline{\mathbf{F}}_q}, \mathbf{Q}_{\ell}(1)) \xrightarrow{\sim} H_{\text{ét}}^{2d-2}(B_{\overline{\mathbf{F}}_q}, \mathbf{Q}_{\ell}(d-1)).$$

Tate's divisor theorem [Tat66, Theorem 4, pp. 143–144] gives surjectivity in codimension one. Multiplication by ϑ^{d-2} gives surjectivity in codimension $d-1$. The unit and the origin generate codimensions zero and d . Together with Broe's theorem, this covers every codimension for $d \leq 4$. $\square$

Applying Theorem 4.1 over every finite extension gives its geometric form: every Tate class on an abelian variety of dimension at most four over $\overline{\mathbf{F}}_p$ is algebraic.

Corollary 4.2. *Let A be an abelian variety of dimension at most four over an algebraically closed field k of characteristic $p > 0$.*

- (i) *For every $\ell \neq p$, numerical and ℓ -adic homological equivalence agree on $\text{CH}^r(A)_{\mathbf{Q}}$ for every r .*
- (ii) *The Lefschetz, Künneth, and Hodge standard conjectures hold for A .*

Proof. For (i) in dimension four, Broe's theorem [Bro26, Corollary 1.2] applies over arbitrary fields for every invertible prime ℓ . If $\dim A = d < 4$, choose an elliptic curve E/k and apply Broe's result to $A \times E^{4-d}$. The zero section $i : A \hookrightarrow A \times E^{4-d}$ and projection π satisfy $\pi_* i_* = 1$ on Chow groups. Both maps preserve numerical and homological equivalence, so D_{ℓ} descends to A . For (ii), the Lefschetz and Künneth standard conjectures follow from the motivic Lefschetz isomorphisms and Chow–Künneth projectors [Anc21, Theorem 4.1(1),(4), p. 14]. Ancona's theorem gives the numerical Hodge standard conjecture in dimension four [Anc21, Theorem 3.18]; in lower dimensions it reduces to the Hodge index theorem. Assertion (i) identifies cycles modulo numerical equivalence with their cohomology classes and identifies the primitive intersection forms. $\square$

5. FROBENIUS TORI AND THE STABILIZER CRITERION

We use the following geometric form of Theorem 1.1.

Theorem 5.1. *Let $A/\overline{\mathbf{F}}_p$ be an abelian variety of dimension at most four. For every $n \geq 1$ and every prime $\ell \neq p$, every Tate class on A^n is algebraic.*

Normalized Frobenius. Pass to a finite constant extension over which all geometric endomorphisms are defined, $q = Q^2$, and the group generated by normalized Frobenius values is torsion-free. If π is Frobenius, put

$$\varphi = Q^{-1}\pi.$$

Then φ has multiplier one. Endomorphisms on each isotypic power give all matrix units. Representatives of the distinct geometric simple isogeny types together generate the tensor category of A , with one Lefschetz coordinate for each distinct root pair; see Remark 5.3(b)–(c).

Let $L = L(A)$ be the Lefschetz group, the centralizer of $\text{End}^0(A)$ in the group of symplectic similitudes of $H^1(A)$. Let $w(\mathbf{G}_m) \subset L$ be the subgroup of scalar homotheties. Write $\nu : L \rightarrow \mathbf{G}_m$ for the multiplier, so that $\nu(w(t)) = t^2$, and put $L' = \ker \nu$. Let $P = P(A)$ be the smallest algebraic subgroup containing a positive power of Frobenius, as in [Mil22, Section 2(c), p. 11]. Put $P' = \ker(\nu|_P)$ and

$$P_0 = \overline{\langle \varphi \rangle}^{\text{Zar}} \subset L'.$$

The constant extension makes P_0 connected: its component group is detected by the torsion among character values of φ . We have

$$\begin{aligned} P &= w(\mathbf{G}_m)P_0, & P' &= \langle [-1] \rangle P_0, \\ \text{im}(P' \rightarrow L'/\langle [-1] \rangle) &= \text{im}(P_0 \rightarrow L'/\langle [-1] \rangle). \end{aligned} \tag{5.1}$$

For the first equality, work over a splitting field. Every character value of φ has complex absolute value one. A relation $Q^a \chi(\varphi) = 1$ therefore forces $a = 0$; the remaining relation is trivial on P_0 by its definition. Thus (Q, φ) is Zariski dense in $\mathbf{G}_m \times P_0$, whose image under multiplication is P . Intersecting with multiplier one proves the second equality, and the third follows on taking the quotient.

The relation lattice. Suppose that L' is a torus, and let Ω split it. Define the *reduced normalized relation lattice* and normalized Frobenius group by

$$\begin{aligned} \text{ev}_A : X^*(L'_\Omega) &\longrightarrow \overline{\mathbf{Q}}^\times, & \chi &\longmapsto \chi(\varphi)^2, \\ \Lambda_A^{\text{red}} &= \ker(\text{ev}_A), & \Gamma'_A &= \text{im}(\text{ev}_A). \end{aligned} \quad (5.2)$$

The chosen constant extension makes Γ'_A torsion-free, so Λ_A^{red} is saturated. The angle rank is $\text{rk } \Gamma'_A$. We call A *neat* when $P(A) = L(A)$; when L' is a torus this is equivalent to $\Lambda_A^{\text{red}} = 0$.

For a simple A with commutative endomorphism algebra of degree $2g$, the lattice has the following coordinates. Over Ω choose the weights $\eta_1, \dots, \eta_g$ occurring on one member of each conjugate pair in $H^1(A)$ and the corresponding Frobenius roots α_i . These weights form a basis of $X^*(L'_\Omega)$, and

$$\begin{aligned} \lambda_i &= \eta_i(\varphi)^2 = \alpha_i^2/q, \\ \Lambda_A^{\text{red}} &= \ker \left(\mathbf{Z}^g \longrightarrow \langle \lambda_1, \dots, \lambda_g \rangle, (m_i) \longmapsto \prod_i \lambda_i^{m_i} \right). \end{aligned} \quad (5.3)$$

Two Weil numbers determine the same *Weil germ* when suitable positive powers are Galois conjugate. The groups and kernels in (5.2) are identified under each further finite constant extension by the injective power map.

Even characters. Consider the cohomology groups of weight zero

$$\mathcal{H}_0(A) = \bigoplus_{n \geq 1} \bigoplus_{r=0}^{n \dim A} H^{2r}(A^n, \mathbf{Q}_\ell(r)).$$

Scalar homotheties act trivially on these spaces, and the faithful Lefschetz action is through

$$\bar{L}' = L/w(\mathbf{G}_m) \simeq L'/\langle [-1] \rangle, \quad \bar{P}' = \text{im}(P_0 \rightarrow \bar{L}').$$

To check faithfulness, a polarization identifies the realization of the Künneth summand $h^1(A) \otimes h^1(A)(1)$ of $h^2(A^2)(1)$ with the endomorphism space of $H^1(A)$. The conjugation action on this space has scalar kernel. The definition of $\bar{P}'$ agrees with the image of P' by (5.1). Pullback identifies its ambient character group with

$$X^*(\bar{L}') = X^*(L')_{\text{ev}} := \{ \chi \in X^*(L') : \chi([-1]) = 1 \}.$$

Thus L' removes the multiplier, $\bar{L}'$ removes the central sign, and $X^*(\bar{L}')$ consists of the even characters of L' . The $\bar{P}'$ -invariants in $H^{2r}(A^n)(r)$ are the Tate classes. The constructed weights will generate the kernel of the restriction map on characters from $\bar{L}'$ to $\bar{P}'$.

Lemma 5.2 (Reduced relations and even characters). *Assume that $\bar{L}'$ is a torus, let Ω split it, and let $\bar{\varphi}$ be the image of normalized Frobenius in $\bar{L}'$.*

(i) *The lattice*

$$\begin{aligned} R_{\text{ev}} &:= \ker[X^*(\bar{L}'_{\Omega}) \longrightarrow X^*(\bar{P}'_{\Omega})] \\ &= \{\chi \in X^*(\bar{L}'_{\Omega}) : \chi(\bar{\varphi}) = 1\} \end{aligned} \quad (5.4)$$

is saturated.

(ii) *If L' is a torus, pullback along $L' \rightarrow \bar{L}'$ gives*

$$R_{\text{ev}} = \Lambda_A^{\text{red}} \cap X^*(L'_{\Omega})_{\text{ev}}. \quad (5.5)$$

Proof. By (5.1), $\bar{P}'$ is the image of the connected torus P_0 ; it is the connected Zariski closure of $\bar{\varphi}$ in $\bar{L}'$. Consequently a character of $\bar{L}'$ restricts trivially to $\bar{P}'$ precisely when its value at $\bar{\varphi}$ is one. This proves (5.4). Restriction of characters to a subtorus is surjective with saturated kernel, proving (i).

If L' is a torus, pullback along $L' \rightarrow \bar{L}'$ identifies $X^*(\bar{L}')$ with the characters of L' trivial on $[-1]$. The normalized Frobenius values generate a torsion-free group, so $\chi(\varphi)^2 = 1$ is equivalent to $\chi(\varphi) = 1$. Intersecting with the even character lattice proves (5.5). $\square$

Remark 5.3 (Coordinates for products). Work over the constant extension fixed after Theorem 5.1. All endomorphisms are geometric.

- (a) For a simple B whose endomorphism algebra is a CM field of degree $2 \dim B$, the coordinates are given by (5.3). A character (m_i) is even if $\sum_i m_i \equiv 0 \pmod{2}$.
- (b) If $A \sim \prod_{\nu} B_{\nu}^{a_{\nu}}$, where the B_{ν} are geometrically simple, nonsupersingular, and pairwise nonisogenous, then

$$X^*(L'_A) = \bigoplus_{\nu} X^*(L'_{B_{\nu}}), \quad (\chi_{\nu})_{\nu} \mapsto \prod_{\nu} \chi_{\nu}(\varphi_{B_{\nu}})^2 \quad (5.6)$$

The map has kernel Λ_A^{red} . In particular, B and B^a have the same coordinates.

- (c) Let B be geometrically simple and nonsupersingular. If $D = \text{End}^0(B)$ has centre K , $d^2 = [D : K]$, and $d[K : \mathbf{Q}] = 2 \dim B$, its coordinates are indexed by conjugate pairs of embeddings of K . Each Frobenius root has multiplicity d .
- (d) Let B be a product of nonsupersingular geometrically simple factors and E_0 a supersingular elliptic curve with normalized Frobenius one. Write e for the nontrivial character of $L'(E_0) = \mu_2$. Then $2e = 0$, and $X^*(\bar{L}'(B \times E_0))$ consists of the pairs $\chi + be$ with $\chi([-1]) = (-1)^b$. If $\Lambda_B^{\text{red}} = \mathbf{Z}u$ and $u([-1]) = -1$, the even kernel for $B \times E_0$ is the free cyclic subgroup of $(\mathbf{Z}u \oplus \mathbf{Z}e)/(2e)$ given by

$$\{\overline{au + be} : a \equiv b \pmod{2}\} = \mathbf{Z}\overline{u + e}, \quad 2\overline{u + e} = \overline{2u}. \quad (5.7)$$

Verification of Remark 5.3. For (a), the weights η_i form a basis and satisfy $\eta_i([-1]) = -1$ and $\eta_i(\varphi)^2 = \alpha_i^2/q$.

For (b), $\text{End}^0(B^a) = M_a(\text{End}^0(B))$. Its matrix units force the centralizer on $H^1(B)^a$ to be the diagonal copy of the centralizer on $H^1(B)$, and Frobenius is the corresponding diagonal element. Thus $L'(B^a) = L'(B)$ and $P'(B^a) = P'(B)$. For pairwise nonisogenous factors the endomorphism algebra and H^1 split as products, whence $L'_A = \prod_{\nu} L'_{B_{\nu}}$, and $P_{0,A}$ is the Zariski closure of the normalized Frobenius tuple $(\varphi_{B_{\nu}})_{\nu}$. Taking characters gives (5.6).

For (c), split D over Ω . The summand for an embedding $K \hookrightarrow \Omega$ is the standard d -dimensional module for $D \otimes_{K,\sigma} \Omega \simeq M_d(\Omega)$. Its commutant consists of scalars. Thus the d equal Frobenius eigenvalues give one Lefschetz character, and complex conjugation pairs the two embeddings.

For (d), the special Lefschetz action on $H^1(E_0)$ is the central sign. A tensor labelled $\chi + be$ descends through the diagonal sign exactly when $\chi([-1]) = (-1)^b$; two copies of e contract by a

polarization. For $\chi = au$, the parity condition $b \equiv a \pmod{2}$ identifies the class with $a\overline{(u+e)}$. Every such multiple satisfies the condition, proving (5.7). $\square$

A good coefficient prime. For a smooth projective variety, the *folklore conjecture* at ℓ asserts that numerical and ℓ -adic homological equivalence coincide. Clozel's theorem gives every abelian variety $A/\overline{\mathbf{F}}_p$ a set $s(A)$ of primes of positive density for which the folklore conjecture holds. In Milne's formulation, $s(A)$ depends only on the simple isogeny factors, so $s(A^n) = s(A)$ [Mil22, Theorem 2.2 and the following paragraph, p. 10]. We call these primes good for A .

Proposition 5.4 (A stabilizer criterion for finitely many cycles). *Let $A/\overline{\mathbf{F}}_p$ be an abelian variety and choose a good prime $\ell_0 \in s(A)$. Suppose that the classes of finitely many cycles $Z_i \in \mathrm{CH}^{r_i}(A^{n_i})_{\mathbf{Q}}$ have common stabilizer $\bar{P}'_{\mathbf{Q}_{\ell_0}}$ in $\bar{L}'_{\mathbf{Q}_{\ell_0}}$. For every $n \geq 1$ and every $\ell \neq p$:*

- (i) *every Tate class on A^n is algebraic;*
- (ii) *numerical and ℓ -adic homological equivalence agree on A^n .*

Proof. *A good prime for all powers.* Since $s(A^n) = s(A)$, numerical and ℓ_0 -adic homological equivalence agree on every A^n [Mil22, Theorem 2.2 and the following paragraph, p. 10].

Algebraic classes and invariant tensors. Let $\mathcal{C}$ be the category of numerical motives generated by $h^1(A)$ and the Tate motive, with coefficients in $\mathbf{Q}_{\ell_0}$ and the symmetry modified by cohomological parity. Jannsen's semisimplicity theorem [Mil22, Theorem A.4, p. 21] and equality of numerical and ℓ_0 -adic homological equivalence on every power make $\mathcal{C}$ semisimple with faithful ℓ_0 -adic realization. Tannakian duality identifies its invariant tensors with algebraic classes [Mil22, Appendix A(a), p. 20]. Let $\bar{M}' \subset \bar{L}'_{\mathbf{Q}_{\ell_0}}$ fix all algebraic classes on powers of A . Algebraic classes are Tate, so

$$\bar{P}'_{\mathbf{Q}_{\ell_0}} \subseteq \bar{M}' \subseteq \mathrm{Stab}_{\bar{L}'_{\mathbf{Q}_{\ell_0}}}((Z_i)_i) = \bar{P}'_{\mathbf{Q}_{\ell_0}}.$$

The inverse image $M \subset L_{\mathbf{Q}_{\ell_0}}$ of $\bar{M}'$ is the Tannakian group of $\mathcal{C}$. Scalar homotheties act trivially on $H^{2r}(A^n, \mathbf{Q}_{\ell_0}(r))$, so its M -invariants are precisely the $\bar{P}'$ -invariants. They are therefore algebraic.

Passage to every prime. The Tate and folklore conjectures now hold on every power at ℓ_0 . Applying [Mil22, Theorem 2.4, p. 11] to each A^n proves both assertions for every $\ell \neq p$. $\square$

Corollary 5.5. *Fix a good prime $\ell_0 \in s(A)$. Assume that $\bar{L}'$ is a torus, and let $\Omega/\mathbf{Q}_{\ell_0}$ split it. Suppose the saturated kernel*

$$R_{\mathrm{ev}} = \ker[X^*(\bar{L}'_{\Omega}) \rightarrow X^*(\bar{P}'_{\Omega})]$$

is generated by $\chi_1, \dots, \chi_m$. Suppose that, for each i , there is a cycle $Z_i \in \mathrm{CH}^{r_i}(A^{n_i})_{\mathbf{Q}}$ whose class over Ω has a nonzero component of weight χ_i . Then every power of A satisfies the Tate conjecture for every $\ell \neq p$.

Proof. A torus element fixing a tensor fixes each of its nonzero weight components. The common stabilizer of the classes of the Z_i therefore lies in $\bigcap_i \ker \chi_i = \bar{P}'$. Since each Z_i is algebraic, this stabilizer also contains $\bar{P}'$. Proposition 5.4 proves the assertion.

The hypothesis can also be checked in the Ω -span of rational cycle classes: if projection onto a weight space is nonzero on that span, it is nonzero on the class of at least one rational cycle. $\square$

Integral generation. The generators in Corollary 5.5 must span R_{ev} over $\mathbf{Z}$. A proper sublattice of finite index gives a common kernel whose quotient by $\bar{P}'$ is a nontrivial finite group. For example, the characters $t \mapsto t$ and $t \mapsto t^2$ of $\mathbf{G}_m$ span the same rational character space, while their kernels are 1 and μ_2 . The ambient lattice for saturation is $X^*(\bar{L}')$. In Proposition 10.5, R_{ev} has index two in the full normalized relation lattice and is saturated in $X^*(\bar{L}')$ by Lemma 5.2.

6. GEOMETRICALLY SIMPLE FOURFOLDS

Lemma 6.1 (A primitive relation with full support). *Let $A/\mathbf{F}_q$ be a geometrically simple abelian fourfold with $\text{End}^0(A_{\overline{\mathbf{F}}_p}) = K$ a CM field of degree eight. Assume that all geometric endomorphisms are defined over $\mathbf{F}_q$, that $q = Q^2$, that $\mathbf{Q}(\pi^N) = K$ for every $N \geq 1$, and that the group generated by the normalized Frobenius roots is torsion-free. In the coordinates of (5.3), either $\Lambda_A^{\text{red}} = 0$ or*

$$\Lambda_A^{\text{red}} = \mathbf{Z}\varepsilon, \quad \varepsilon \in \{\pm 1\}^4. \quad (6.1)$$

Proof. Choose the indexing of Frobenius roots in [APFV25, Definition 2.3]. Geometric simplicity and the stated endomorphism field verify [APFV25, Definition 1.10]. By the classification in [APFV25, Appendix A.1, Lemma A.1 and Tables A.3–A.6], the angle rank is four or three, and every entry of rank three is exceptional. Thus the reduced relation lattice is zero or has rank one.

For angle rank three, [APFV25, Lemmas 4.2–4.3] give disjoint nonempty sets $T^+, T^- \subset \{1, 2, 3, 4\}$ of even total cardinality and

$$\prod_{i \in T^+} (\alpha_i/Q) \prod_{j \in T^-} (\alpha_j/Q)^{-1} = \zeta$$

with ζ a root of unity. The torsion hypothesis gives $\zeta = 1$. The corresponding vector ε has entries in $\{0, \pm 1\}$, both signs, and even support. Support two would make two distinct Frobenius roots have equal positive powers, whereas $\mathbf{Q}(\pi^N) = K$ has degree eight for every $N \geq 1$. The support therefore has size four. Since ε is primitive and the relation lattice is saturated of rank one, (6.1) follows. $\square$

Theorem 6.2. *If $A/\overline{\mathbf{F}}_p$ is a geometrically simple abelian fourfold, the Tate conjecture holds for every power of A and every $\ell \neq p$.*

Proof. Choose a finite field over which all geometric endomorphisms are defined, every positive power of Frobenius generates the centre of the geometric endomorphism algebra, and normalized Frobenius values generate a torsion-free group. Fix a good prime $\ell_0 \in s(A)$ and a splitting field $\Omega/\mathbf{Q}_{\ell_0}$ for its Lefschetz torus.

Commutative endomorphism algebra. Suppose that $K = \text{End}^0(A) = \mathbf{Q}(\pi)$ is a CM field of degree eight. Lemma 6.1 gives $\Lambda_A^{\text{red}} = 0$ or (6.1).

If $\Lambda_A^{\text{red}} = 0$, then A is neat. Every Tate tensor on a power of A is a Lefschetz tensor and is algebraic by [Mil22, Theorem 2.5]. Otherwise, write

$$H^1(A, \Omega) = \bigoplus_{i=1}^4 (V_i \oplus V_i).$$

The two weight lines

$$W_{\pm\varepsilon} = \bigwedge_{\pm\varepsilon_i=1} V_i \wedge \bigwedge_{\pm\varepsilon_i=-1} V_i \subset H^4(A, \Omega)(2) \quad (6.2)$$

are fixed by Frobenius. By Theorem 4.1, both lines lie in the Ω -span of rational cycle classes. Equations (5.5) and (6.1) give $R_{\text{ev}} = \mathbf{Z}\varepsilon$. Corollary 5.5 proves Tate on every power of A when $\text{End}^0(A)$ is commutative.

Noncommutative endomorphism algebra. Put

$$D = \text{End}^0(A), \quad K = Z(D) = \mathbf{Q}(\pi), \quad d^2 = [D : K], \quad e = [K : \mathbf{Q}].$$

The dimension formula from [Tat66, Theorem 2(a),(b)], recalled in [Zar15, Example 2.4], gives

$$de = 2 \dim A = 8.$$

If π is real, A is supersingular. Otherwise K is CM, and the noncommutative possibilities are $(d, e) = (2, 4)$ and $(4, 2)$.

All geometric endomorphisms are defined over the chosen field, so

$$\mathrm{End}_{\mathbf{F}_{q^N}}^0(A) = D, \quad \mathbf{Q}(\pi^N) = K \quad (N \geq 1). \quad (6.3)$$

Hence the quotient of two distinct K -conjugates of π has infinite multiplicative order. Write $e = 2m$. By Remark 5.3(c), the reduced normalized relation lattice is a saturated sublattice of $\mathbf{Z}^m$ stable under Galois, with Galois acting transitively by signed permutations. If $m = 1$, a nonzero relation makes the normalized root torsion, contrary to (6.3). If $m = 2$, a relation lattice of rank two makes both normalized roots torsion. A stable lattice of rank one has a primitive generator (a, b) ; a signed permutation exchanges the two coordinates and preserves the line, hence $|a| = |b| = 1$. Either $(1, 1)$ or $(1, -1)$ again makes a quotient of two distinct conjugates torsion. Thus $\Lambda_A^{\mathrm{red}} = 0$, and A is neat. The Tate conjecture holds on every power by [Mil22, Theorem 2.5]. A supersingular A is geometrically isogenous to a product of elliptic curves, so Spieß's theorem applies [Mil22, Theorem 4.1(c)]. $\square$

7. MIXED ISOGENY TYPES

By Remark 5.3(b)–(d), it suffices to treat the geometric isogeny types

$$3 + 1, \quad 2 + 2, \quad 2 + 1 + 1, \quad 1 + 1 + 1 + 1.$$

The last type follows from Spieß's theorem [Mil22, Theorem 4.1(c)].

7.1. The type $3 + 1$. Write $A \sim X \times E$, with X a geometrically simple threefold and E an elliptic curve. If X is neat, [Mil22, Theorem 4.2] proves Tate for $X^r \times E^s$ for all r, s .

Suppose that X is exceptional. By Zarhin's classification [Zar15, Theorem 1.1 and the proof of Corollary 4.6], over the chosen finite field X is absolutely simple and almost ordinary, $E_X = \mathrm{End}^0(X) = \mathbf{Q}(\pi_X)$ is a sextic CM field, and its angle rank is two. Moreover, E_X contains an imaginary quadratic field B such that

$$\mathrm{Nm}_{E_X/B}(\pi_X^2/q) = 1. \quad (7.1)$$

The field B is unique: two distinct quadratic subfields would generate a quartic subfield of the sextic field E_X . Choose the three embeddings of E_X above one embedding of B . The norm identity (7.1) gives the primitive reduced relation $u = (1, 1, 1)$ with full support. We first construct a cycle of weight $2u$ on X^2 .

Lemma 7.1. (i) *For some good prime $\ell_0 \in s(X)$, there is a cycle $\Xi \in \mathrm{CH}^3(X^2)_{\mathbf{Q}}$ whose class over a splitting field of $L'(X)_{\mathbf{Q}_{\ell_0}}$ has a nonzero component of weight $2u$.*

(ii) *For every $n \geq 1$ and every $\ell \neq p$, every Tate class on X^n is algebraic.*

Proof. A class on a fourfold. Choose a good prime ℓ_0 for X . After a finite constant extension, normalized torsion is trivial and there is a supersingular elliptic curve E_0 with scalar Frobenius Q , where $q = Q^2$. Enlarge the field so that all geometric endomorphisms of E_0 are defined. Over a splitting field $\Omega/\mathbf{Q}_{\ell_0}$, let $W_u \subset H^3(X, \Omega)$ be a line of weight u . The relation $u = (1, 1, 1)$ gives Frobenius Q^3 on W_u after the chosen sign has been killed. Hence Frobenius on $W_u \otimes H^1(E_0, \Omega)$ is $Q^4 = q^2$, and

$$W_u \otimes H^1(E_0, \Omega) \subset H^4(X \times E_0, \Omega)(2)$$

is Tate. By Theorem 4.1, the classes of rational cycles span the Tate space of $X \times E_0$ after scalar extension. Choose $Z \in \mathrm{CH}^2(X \times E_0)_{\mathbf{Q}}$ whose class has nonzero projection $w \otimes v$ to $W_u \otimes H^1(E_0, \Omega)$, with $w \neq 0$ and $v \neq 0$. Apply the rational Chow–Künneth projector onto $h^3(X) \otimes h^1(E_0)$ to Z . In this Künneth summand the Tate classes have X -weights u and $-u$: these are the only weights in $\mathbf{Z}u$ occurring in $H^3(X)$.

Contraction along the elliptic factors. Let ψ be a polarization of E_0 . Tate's isogeny theorem [Tat66, Main Theorem, §1] gives

$$\mathrm{End}^0(E_0) \otimes_{\mathbf{Q}} \Omega = \mathrm{End}_{\Omega}(H^1(E_0, \Omega)).$$

The linear functional $a \mapsto \psi(v, av)$ on this endomorphism algebra is nonzero. It is therefore nonzero on some $a \in \mathrm{End}^0(E_0)$; clearing denominators makes a an endomorphism of E_0 . Put $Z_a = (1_X \times a)^*Z$. Form $Z \boxtimes Z_a$ and contract the two $h^1(E_0)$ factors by the polarization $h^1(E_0) \otimes h^1(E_0) \rightarrow \mathbf{L}$. This gives a rational cycle $\Xi \in \mathrm{CH}^3(X^2)_{\mathbf{Q}}$. Its component of weight $2u$ is a nonzero scalar multiple of

$$\psi(v, av) w \otimes w.$$

Only the two components of X -weight u contribute to weight $2u$.

All powers of the threefold. The reduced relation lattice of X is $\mathbf{Z}u$, and Lemma 5.2 identifies its even character kernel with $2\mathbf{Z}u$. Corollary 5.5 applied to $2u$ proves (ii). $\square$

Proposition 10.4 uses this elliptic contraction for a dihedral pair, together with a contraction on an abelian surface. The resulting weights generate the even relation lattice by Proposition 10.5.

An ordinary elliptic factor. Suppose that the given E is ordinary and put $K = \mathrm{End}^0(E)$. If its normalized Frobenius group met that of X nontrivially, then, since their ranks are one and two, the product would have rank $\mathrm{rk} \Gamma'_X + \mathrm{rk} \Gamma'_E - 1 = 2$. Zarhin's Theorem 3.1 would supply an imaginary quadratic field B_0 and embeddings $B_0 \hookrightarrow E_X$ and $B_0 \hookrightarrow K$ for which $\mathrm{Nm}_{E_X/B_0}(\pi_X^2/q)$ has infinite multiplicative order [Zar15, Theorem 3.1]. The uniqueness of the imaginary quadratic subfield of E_X , and the fact that K itself is quadratic, identify these copies with B and K ; this contradicts (7.1). Thus the groups are independent. Remark 5.3(b) therefore identifies the mixed even kernel with the even part of $\mathbf{Z}u$, namely $2\mathbf{Z}u$. The Tate conjecture for every mixed power now follows from Lemma 7.1, the neatness of E [Mil22, Theorem 4.1(c)], and Corollary 5.5.

A supersingular elliptic factor. If E is supersingular, make its normalized Frobenius scalar and denote its odd character by e . By Remark 5.3(d), modulo divisor pairings the relevant symbols lie in $(\mathbf{Z}u \oplus \mathbf{Z}e)/(2e)$, and their even kernel is exactly

$$\{\overline{au + be} : a \equiv b \pmod{2}\} = \mathbf{Z}\overline{u + e}. \quad (7.2)$$

This generator occurs already on the fourfold $X \times E$ and is algebraic by Theorem 4.1. Its double is $2u$, as in (5.7). Thus Corollary 5.5 proves the Tate conjecture for every mixed power with supersingular E .

7.2. The type 2+1+1. Write $A \sim S \times E_1 \times E_2$. If S is geometrically decomposable or supersingular, this reduces to elliptic curves. Otherwise S is geometrically simple; like every abelian surface over $\mathbf{F}_p$, it is neat [Mil22, §4(b)].

The quartic endomorphism field. The dimension formula in [Zar15, Example 2.4] leaves a quartic CM field or a quaternion division algebra over an imaginary quadratic center. For a quaternion division algebra, the local invariant formula [Tat66, §3, formula (7), p. 142] forces p to split in the center and both slopes to equal $1/2$: a nonsplit prime has slope $1/2$, local degree two, and invariant zero, whereas a split prime has invariant equal to its slope. This would make S supersingular. Consequently $F = \mathrm{End}^0(S) = \mathbf{Q}(\pi_S)$ is a quartic CM field. It is primitive and contains no imaginary quadratic subfield by [Zar15, Corollary 5.2]; the hypothesis there that all endomorphisms be defined over the finite base field holds by the constant extension in Section 5.

Independence from elliptic factors. For every ordinary E_i , the groups of S and E_i are independent: a rank-one intersection would force a common imaginary quadratic subfield by [Zar15, Theorem 3.1]. Supersingular elliptic factors have scalar normalized Frobenius after the chosen constant extension and are handled by Remark 5.3(d). To verify independence for all three factors, combine isogenous elliptic factors first. For two distinct ordinary elliptic types, put $K_i = \mathrm{End}^0(E_i)$; their fields are distinct: for ordinary elliptic curves with the same quadratic Frobenius field, the Frobenius ideals

agree up to conjugation. The quotient of the corresponding roots is then a unit whose complex absolute values are one, hence a root of unity. Equal powers give an isogeny by [Tat66, Main Theorem, §1]. Let L be the normal closure of F . Its Galois group is a transitive subgroup of S_4 with central complex conjugation, so it is C_4 , D_4 , or V_4 . The last possibility would give an imaginary quadratic subfield of F . In the first two groups complex conjugation is the central square, so every quadratic subfield of L is real. Choose a generator γ_i of the infinite cyclic group Γ'_{E_i} . A relation

$$x\gamma_1^a\gamma_2^b = 1, \quad x \in \Gamma'_S, \quad a, b \in \mathbf{Z},$$

puts x in $L \cap K_1 K_2$, which is $\mathbf{Q}$ or a real quadratic field. All conjugates of x have absolute value one, so $x = \pm 1$; normalized torsion has been killed, hence $x = 1$. The remaining relation between the two nonisogenous ordinary elliptic factors then has $a = b = 0$. Supersingular elliptic factors have scalar normalized Frobenius after the chosen extension. Since the reduced normalized relation lattice of S and all ordinary elliptic factors is zero by Remark 5.3(b), any relation involving a supersingular symbol has zero nonsupersingular part; its even part is a pure elliptic Lefschetz relation, covered by Spieß's theorem [Mil22, Theorem 4.1(c)]. Lemma 5.2 and Corollary 5.5 prove the assertion for all powers of the type $2 + 1 + 1$.

7.3. Reduction of the type $2 + 2$. Let $A \sim S \times T$. Decomposable or supersingular surfaces are covered by Section 7.2. Every abelian surface is neat [Mil22, Section 4(b)], so geometrically isogenous factors satisfy the Tate conjecture on all powers. Assume that S, T are geometrically simple and nonisogenous. Their endomorphism algebras are primitive quartic CM fields by the paragraph *The quartic endomorphism field* in Section 7.2. Thus their special Lefschetz tori have dimension two and $\text{rk } \Gamma'_S = \text{rk } \Gamma'_T = 2$. Put

$$d(S, T) = \text{rk}(\Gamma'_S \cap \Gamma'_T) \in \{0, 1, 2\}.$$

Equivalently, $\text{rk } \Gamma'_{S \times T} = 4 - d(S, T)$. If $d = 0$, Remark 5.3(b) shows that the product is neat. If $d = 1$, Zarhin's Theorem 3.1 produces a common imaginary quadratic subfield, while the primitive quartic centres exclude such a field [Zar15, Theorem 3.1 and Corollary 5.2]. Thus $d = 2$ is the remaining possibility:

$$d(S, T) = 2, \quad \Gamma'_S \text{ and } \Gamma'_T \text{ commensurable.} \quad (7.3)$$

Lemma 8.1 classifies the pairs satisfying (7.3). Theorem 10.6 proves the Tate conjecture for all their powers.

8. DIHEDRAL PAIRS OF ABELIAN SURFACES

Lemma 8.1. *Let $S, T/\overline{\mathbf{F}}_p$ be geometrically simple, nonisogenous abelian surfaces, with models over one sufficiently large finite field $\mathbf{F}_q$. Suppose*

$$\text{rk } \Gamma'_S = \text{rk } \Gamma'_T = 2, \quad \Gamma'_S \sim \Gamma'_T, \quad (8.1)$$

where $\sim$ denotes commensurability. Then:

- (i) *the fields $K_S = \mathbf{Q}(\pi_S)$ and $K_T = \mathbf{Q}(\pi_T)$ are primitive quartic CM fields with a common normal closure L and $\text{Gal}(L/\mathbf{Q}) \simeq D_4$;*
- (ii) *the decomposition group D of a prime of L above p is a reflection;*
- (iii) *the two centres are fixed by reflections in different conjugacy classes;*
- (iv) *after interchanging S, T , their Newton slopes are $(0, \frac{1}{2}, \frac{1}{2}, 1)$ and $(0, 0, 1, 1)$, respectively;*
- (v) *for fixed (L, D) , each reflection conjugacy class gives at most one geometric isogeny class of geometrically simple surfaces with normalized Frobenius rank two and centre fixed by a reflection in that class.*

Conversely, suppose that S, T have Frobenius fields fixed by reflections in different conjugacy classes of a common D_4 extension $L/\mathbf{Q}$, and that a decomposition group of L above p is a reflection. If both normalized Frobenius groups have rank two, then they are commensurable.

Proof. The common normal closure. For a simple surface X , Honda–Tate theory gives $d[K_X : \mathbf{Q}] = 4$, where $d^2 = [\text{End}^0(X) : K_X]$ [Wat69, Chapter 2, pp. 526–528]. A quadratic centre gives at most one reduced coordinate by Remark 5.3(c), and a real centre is supersingular. Thus rank two forces $d = 1$ and $[K_X : \mathbf{Q}] = 4$. By [Zar15, Corollary 5.2], K_X contains no imaginary quadratic subfield. It is therefore primitive, and its normal closure has group C_4 or D_4 .

Let L_S, L_T be these normal closures and put $M = L_S \cap L_T$. The intersection $\Gamma'_S \cap \Gamma'_T \subset M^\times$ has rank two, and all conjugates of each of its elements have absolute value one. Hence M has a nonreal embedding. Every proper Galois subfield of either L_X is totally real: in C_4 the unique subgroup of order two is complex conjugation, and every nontrivial normal subgroup of D_4 contains complex conjugation. Since $M/\mathbf{Q}$ is Galois, $L_S = M = L_T =: L$.

Put $G = \text{Gal}(L/\mathbf{Q})$, let c be complex conjugation, and choose a prime $\mathfrak{p} \mid p$ with decomposition group D . For $x \in L^\times$ whose finite divisor is supported above p , define

$$\text{Div}_p(x) = \sum_{gD \in G/D} \frac{v_{g\mathfrak{p}}(x)}{v_{g\mathfrak{p}}(q)} [gD] \in \mathbf{Q}[G/D]. \quad (8.2)$$

This map is G -equivariant. Write $K_X = L^{H_X}$ and $\lambda_X = \pi_X^2/q$. The relation $c\lambda_X = \lambda_X^{-1}$ defines

$$\begin{aligned} V_{H_X} &= \mathbf{Q}[G/H_X]^-, \\ \tilde{d}_{\pi_X, H_X} : \mathbf{Q}[G/H_X] &\longrightarrow \mathbf{Q}[G/D]^-, \\ [gH_X] &\longmapsto \text{Div}_p(g\lambda_X), \\ d_{\pi_X, H_X} &= \tilde{d}_{\pi_X, H_X}|_{V_{H_X}}. \end{aligned} \quad (8.3)$$

The map is well defined because H_X fixes λ_X ; it kills the c -invariant part, and its image is

$$\text{Div}_p(\Gamma'_X) \otimes_{\mathbf{Z}} \mathbf{Q}. \quad (8.4)$$

Every element of the joint normalized Frobenius group is a unit away from p and has absolute value one at each archimedean place. If its p -divisor vanishes, Kronecker's theorem makes it a root of unity. Our constant extension has killed this torsion. Thus Div_p is injective on the joint group, each map d_{π_X, H_X} has rank two, and

$$\Gamma'_S \sim \Gamma'_T \iff \text{im } d_{\pi_S, H_S} = \text{im } d_{\pi_T, H_T} \text{ in } \mathbf{Q}[G/D]^-. \quad (8.5)$$

Equality of the rational images gives an intersection of finite index after clearing denominators, since the divisor map is injective on the joint group. For conjugate normalized principal divisors, injectivity of Div_p identifies the corresponding normalized roots. Tate's homomorphism theorem [Tat66, Main Theorem, §1] then makes the surfaces geometrically isogenous.

The decomposition group. The nonzero image in $\mathbf{Q}[G/D]^-$ forces $c \notin D$. If $G = C_4$, this forces $D = 1$. The local invariant formula in [Wat69, Chapter 2, p. 527] then makes each slope integral, hence zero or one. The four choices of one prime from each conjugate pair form one C_4 -orbit. The normalized principal divisors are therefore conjugate, contradicting nonisogeny.

Now $G = D_4$, and D is either trivial or a reflection. If $D = 1$, the slopes are again zero or one. Their normalized functions are H_X -fixed maps $\epsilon : G \rightarrow \{\pm 1\}$ with $\epsilon(cg) = -\epsilon(g)$. Each of the four reflection subgroups H_X gives four such functions. Appendix A, (A.2)–(A.3), shows that two of these functions have the same rational image plane precisely when they belong to one G -orbit. Equation (8.5) would therefore make their normalized divisors conjugate and their surfaces geometrically isogenous. Thus D is a reflection.

Newton slopes. Write

$$G = \langle r, s : r^4 = s^2 = 1, \text{ } srs = r^{-1} \rangle, \quad c = r^2, \quad D = \langle s \rangle.$$

Every primitive quartic CM subfield is fixed by a reflection H . If H is conjugate to D , the D -orbits on G/H have sizes 1, 1, 2. The singleton slopes are complementary integers, and the orbit of size

two is c -stable, hence has slope $1/2$. If H lies in the other reflection class, the orbit sizes are $2, 2$. Local integrality allows slopes $0, 1/2, 1$ on either orbit, with complementary values on the other. The constant slope $1/2$ gives rank zero. The two reflection classes therefore give, up to ordering and complementation,

$$(0, 1, \tfrac{1}{2}, \tfrac{1}{2}), \quad (0, 0, 1, 1). \quad (8.6)$$

Uniqueness and the converse. Within each reflection class the slope choices form one G -orbit. Injectivity of Div_p therefore gives (v). Nonisogeny forces H_S, H_T into different classes, proving (iii)–(iv).

For the converse, put $f_j = r^j D$, $A = f_0 - f_2$, and $B = f_1 - f_3$. In the basis $e_0 - e_2, e_1 - e_3$ of $\mathbf{Q}[G/H]^-$, where $e_j = r^j H$, the two functions in (8.6) give

$$[d]_{\text{same}} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad [d]_{\text{opposite}} = \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix}. \quad (8.7)$$

Both images are $\mathbf{Q}[G/D]^-$, so (8.5) proves the converse. $\square$

8.1. An explicit dihedral pair.

Proposition 8.2. *There are geometrically simple, nonisogenous abelian surfaces over $\mathbf{F}_{29^2}$ with commensurable normalized Frobenius groups of rank two. Their Frobenius polynomials are*

$$\begin{aligned} P_1(X) &= X^4 + 40X^3 + 1102X^2 + 33640X + 707281, \\ P_2(X) &= X^4 + 20X^3 + 1078X^2 + 16820X + 707281. \end{aligned} \quad (8.8)$$

Proof. Frobenius fields and roots. Let $u = \sqrt{5}$, and choose $\theta^2 = -4 + u$ and $(\theta')^2 = -4 - u$ with $\theta\theta' = \sqrt{11}$. The fields

$$F_1 = \mathbf{Q}(\theta), \quad F_2 = \mathbf{Q}(\theta + \theta')$$

have defining polynomials $X^4 + 8X^2 + 11$ and $X^4 + 16X^2 + 20$. They have a common D_4 normal closure and correspond to the two reflection classes. Put $q = 29^2$. The integral elements

$$\pi_1 = (-10 + 7u) + (8 + 6u)\theta, \quad \pi_2 = (-5 - 4\sqrt{11}) + (6 + 4\sqrt{11})(\theta + \theta')$$

satisfy $\pi_i \bar{\pi}_i = q$ and have characteristic polynomials (8.8). Modulo 3, these are Φ_5 and Φ_{10} , both irreducible, so $\mathbf{Q}(\pi_i) = F_i$.

Newton slopes and isogeny classes. At 29, the defining polynomials have squarefree factorization patterns $(1, 1, 2)$ and $(2, 2)$. The Newton slopes of P_1, P_2 are $(0, \frac{1}{2}, \frac{1}{2}, 1)$ and $(0, 0, 1, 1)$. Their local Brauer invariants vanish, so Honda–Tate theory gives simple surfaces with endomorphism fields F_1, F_2 [Wat69, Chapter 2, pp. 527–528]. Each slope function has stabilizer exactly the reflection fixing F_i ; hence $\mathbf{Q}(\pi_i^n) = F_i$ for every $n \geq 1$, and the surfaces are geometrically simple. Their different Newton polygons exclude geometric isogeny. The decomposition group is a reflection, and each nonzero normalized divisor spans the irreducible two-dimensional module $\mathbf{Q}[G/D]^-$. Equation (8.5) gives the required commensurability. $\square$

9. CURVES GENERATING THE THIRD COHOMOLOGY

We prove Theorem 1.2 by constructing curve correspondences for abelian threefolds with Newton slopes $0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1$. An isogeny pencil gives a positive Gross–Schoen height and a nonzero primitive class. Monodromy generates the primitive cohomology, and specialization reaches every prescribed isogeny class.

9.1. An isogeny pencil through a prescribed point.

Lemma 9.1. *Let (Y, λ_Y) be a principally polarized abelian threefold over an algebraically closed field k of characteristic $p > 0$, with Newton slopes $0, 1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}$ and $a(Y) = 1$. The following hold.*

- (i) *There are a polarized abelian threefold (A_0, λ_0) with $\ker \lambda_0 \simeq \alpha_p^2$ and isogenies*

$$q_0 : A_0 \longrightarrow Y, \quad u_0 : Y \longrightarrow A_0, \quad q_0 u_0 = [p]_Y,$$

such that $\deg q_0 = p$ and $\lambda_0 = q_0^ \lambda_Y$.*

- (ii) *The quotients by the lines in $\ker \lambda_0$ form a principally polarized abelian scheme $(\mathcal{A}, \lambda)/\mathbf{P}^1$ and an isogeny*

$$q : A_0 \times \mathbf{P}^1 \longrightarrow \mathcal{A}$$

such that a fibre of $(\mathcal{A}, \lambda)$ is (Y, λ_Y) . The determinant of the Hodge bundle of $\mathcal{A}/\mathbf{P}^1$ has degree $p - 1$.

- (iii) *The relative isogeny $f = q \circ (u_0 \times 1) : Y \times \mathbf{P}^1 \rightarrow \mathcal{A}$ satisfies*

$$\deg f = p^6, \quad f^* \lambda = p^2 \lambda_{Y \times \mathbf{P}^1}.$$

- (iv) *If (Y, λ_Y) is defined over $k_0 \subset k$, these data may be chosen over a finite extension of k_0 inside k .*

Proof. Fix a field of definition k_0 as in (iv). It suffices to construct the data over the algebraic closure of k_0 inside k . For the lattice calculation, denote this algebraic closure by k and use covariant Dieudonné modules over $W(k)$.

The inverse isogenies. The connected-étale splitting over the perfect field k , and its Cartier dual, split off the height-two ordinary summand. The polarization makes the ordinary summand self-dual and orthogonal to a self-dual supersingular lattice M of height four and dimension two. Put $N = FM + VM$. Then $\text{length}_W(M/N) = a(M) = 1$, and $pM \subset N$. For a lattice $L \subset M[1/p]$, write $L^\vee$ for its dual under the fixed rational alternating pairing.

On $H = M/pM$, the operators F, V are nilpotent of rank two and

$$\ker F = \text{im } V, \quad \ker V = \text{im } F, \quad \dim(\text{im } F \cap \text{im } V) = 1.$$

Since $a(M) = 1$, the subspaces $\text{im } F$ and $\text{im } V = \ker F$ are distinct. Hence $F^2 \neq 0$. The nonzero image spaces of a nilpotent semilinear map strictly decrease. Since $\text{rank } F = 2$ and $F^2 \neq 0$, we have $\text{rank } F^2 = 1$ and $F^3 = 0$. Interchanging F and V gives $\text{rank } V^2 = 1$ and $V^3 = 0$. Hence

$$\text{im } F^2 = \text{im } F \cap \text{im } V = \text{im } V^2.$$

Lifting to M gives

$$FN = F^2 M + pM = V^2 M + pM = VN.$$

Thus N is superspecial. The polarization identities

$$\langle Fx, y \rangle = \sigma \langle x, Vy \rangle, \quad \langle Vx, y \rangle = \sigma^{-1} \langle x, Fy \rangle$$

give the lattice equalities

$$(FL)^\vee = V^{-1} L^\vee, \quad (VL)^\vee = F^{-1} L^\vee.$$

Taking duals of $FN = VN$ and applying $FV = VF = p$ therefore gives $FN^\vee = VN^\vee$. Since $M \subset N^\vee$,

$$N = FM + VM \subset FN^\vee + VN^\vee = FN^\vee.$$

Self-duality of M gives

$$\text{length}_W(N^\vee/M) = \text{length}_W(M/N) = 1, \quad \text{length}_W(N^\vee/N) = 2.$$

For any F, V -stable lattice in $M[1/p]$, the colength of its F -image is the valuation of $\det F$, equal to the sum of its four slopes, namely two. Thus $\text{length}_W(N^\vee/FN^\vee) = 2$ as well. Consequently

$$N = FN^\vee = VN^\vee, \quad N^\vee/N \simeq D(\alpha_p^2).$$

Retain the ordinary lattice and replace M by N . The resulting lattice N_{full} lies between $pD(Y)$ and $D(Y)$. It is realized by a quotient $u_0 : Y \rightarrow A_0$ with kernel in $Y[p]$: in covariant modules the map is $p : D(Y) \rightarrow N_{\text{full}}$. Factoring $[p]$ gives $q_0 : A_0 \rightarrow Y$, whose covariant map is the inclusion $N_{\text{full}} \subset D(Y)$. The colengths are

$$\text{length}_W(N_{\text{full}}/pD(Y)) = 6 - 1 = 5, \quad \text{length}_W(D(Y)/N_{\text{full}}) = 1.$$

Thus u_0 and q_0 have degrees p^5 and p , respectively, and their composite $[p]$ has degree p^6 .

For an isogeny $h : A \rightarrow B$ with $\lambda_A = h^\vee \lambda_B h$, the covariant pairing satisfies

$$\psi_{\lambda_A}(x, y) = \psi_{\lambda_B}(D(h)x, D(h)y).$$

Since $D(q_0)$ is the inclusion $N_{\text{full}} \hookrightarrow D(Y)$, the pairing of $\lambda_0 = q_0^* \lambda_Y$ is the restriction of the pairing on $D(Y)$. Its discriminant module is $N^\vee/N$, so $\ker \lambda_0 \simeq \alpha_p^2$ and $\deg \lambda_0 = p^2$. The map $D(u_0)$ is multiplication by p , and hence $u_0^* \psi_{\lambda_0} = p^2 \psi_{\lambda_Y}$. This proves (i).

The principally polarized quotients. The alternating Witt pairing induces an alternating linking form on $N^\vee/N$ with values in $p^{-1}W/W$. The inverse image in $N^\vee$ of every line $\ell \subset N^\vee/N$ is an integral, self-dual lattice. For $q_\ell : A_0 \rightarrow B = A_0/H_\ell$, the rational homomorphism

$$\mu = (q_\ell^\vee)^{-1} \lambda_0 q_\ell^{-1}$$

therefore induces an integral, perfect Dieudonné pairing on $D(B)$. To see that μ is an actual homomorphism, choose r such that $p^r \mu$ is an actual homomorphism. Integrality implies that $p^r \mu$ kills $B[p^r]$; factoring through $[p^r]_B$ gives $\mu \in \text{Hom}(B, B^\vee)$. Since $q_\ell^* \mu = \lambda_0$ and $\deg \mu = \deg \lambda_0 / (\deg q_\ell)^2 = 1$, polarization descent [vdGM11, Chapter XI, Proposition 11.25(ii), pp. 169–170] makes μ a principal polarization. At the generic point of $\mathbf{P}^1$, construct μ over an algebraic closure and descend μ by its rational formula and faithful flatness. The resulting homomorphism extends between the abelian schemes; its pullback is λ_0 everywhere, so its fibres are principal polarizations. The line M/N recovers (Y, λ_Y) .

The Hodge degree and descent. The universal line subgroup has Lie bundle $L = \mathcal{O}(-1)$. The exact Lie sequence for a height-one quotient [RS22, Theorem 1.2 and (1.3), p. 97] gives

$$0 \longrightarrow L \longrightarrow \mathcal{O}^{\oplus 3} \longrightarrow \text{Lie}(\mathcal{A}/\mathbf{P}^1) \longrightarrow L^{\otimes p} \longrightarrow 0.$$

Its first cokernel is $\mathcal{O}(1) \oplus \mathcal{O}$, and the remaining extension splits because $H^1(\mathbf{P}^1, \mathcal{O}(p+1) \oplus \mathcal{O}(p)) = 0$. Thus

$$\text{Lie}(\mathcal{A}/\mathbf{P}^1) = \mathcal{O}(1) \oplus \mathcal{O} \oplus \mathcal{O}(-p),$$

and the determinant of the Hodge bundle has degree $p-1$. This proves (ii). For each fibre, $f_t = q_t u_0$ and $q_t^* \lambda_t = \lambda_0$, so

$$f_t^* \lambda_t = u_0^* q_0^* \lambda_Y = [p]^* \lambda_Y = p^2 \lambda_Y.$$

The degree is $p \cdot p^5 = p^6$, proving (iii). The construction uses finitely many coefficients in the algebraic closure of k_0 , so all data descend to a finite extension of k_0 . This proves (iv). $\square$

Lemma 9.2 (The separating boundary estimate). *Let $g : \mathcal{C} \rightarrow T$ be a stable family of curves of genus three and compact type over a smooth projective connected curve over an algebraically closed field. Suppose that the generic fibre is smooth and nonhyperelliptic. Put $\lambda = \det g_* \omega_g$, and let δ_1 count separating nodes with their thicknesses. Then $\delta_1 \leq 3 \deg \lambda$.*

Proof. On the stack $\overline{\mathcal{M}}_3$ over the given field, let f be the universal curve and consider

$$\mu : \text{Sym}^2 f_* \omega_f \longrightarrow f_* \omega_f^{\otimes 2}.$$

Both bundles have rank six. The canonical model of a smooth nonhyperelliptic curve of genus three is a plane quartic, so μ is generically invertible.

At the generic point of the separating boundary Δ_1 , choose an étale chart and its discrete valuation ring R , with uniformizer t . The special curve has components of genera one and two. Its

space of differentials is $V_1 \oplus V_2$, with $\dim V_i = i$. The direct summand $V_1 \otimes V_2$ of $\text{Sym}^2(V_1 \oplus V_2)$ maps to zero under multiplication. Thus the matrix of μ modulo t has corank $c \geq 2$. Invertible row and column operations over R put the matrix in the form $\text{diag}(I_{6-c}, tB)$. Consequently

$$\text{ord}_t(\det \mu) \geq c \geq 2.$$

This calculation holds in every characteristic, including characteristic two.

Regularity of $\overline{\mathcal{M}}_3$ makes $\text{div}(\det \mu) - 2\Delta_1$ effective. Its pullback to T is effective because the generic fibre is smooth and nonhyperelliptic. A node with local equation $xy = u^m$ contributes m to the pullback of Δ_1 , so its degree is δ_1 . Grothendieck–Riemann–Roch gives degree $9 \deg \lambda - \delta_1$ for the determinant of the multiplication map on T . The residual effective divisor has degree $9 \deg \lambda - 3\delta_1 \geq 0$. $\square$

Proposition 9.3. *Let (Y, λ_Y) satisfy the hypotheses of Lemma 9.1, and suppose that it is the Jacobian of a nonhyperelliptic curve of genus three. Let $\mathcal{A} \rightarrow \mathbf{P}^1$ be the abelian scheme in Lemma 9.1(ii). There are a finite map $T \rightarrow \mathbf{P}^1$ from a smooth projective connected curve, a stable family $g : \mathcal{C} \rightarrow T$ of curves of genus three and compact type with smooth generic fibre C , and a line bundle ξ of degree one on C with $\xi^{\otimes 4} \simeq \omega_C$, satisfying the following.*

- (i) *The principally polarized Jacobian of $\mathcal{C}/T$ is $\mathcal{A}_T$.*
- (ii) *If $d = \deg(T \rightarrow \mathbf{P}^1)$, $\lambda = \det g_* \omega_g$, and δ_1 is the number of separating nodes counted with thickness, then*

$$\deg \lambda = d(p-1), \quad \langle \Delta_\xi, \Delta_\xi \rangle = 21 \deg \lambda - 6\delta_1 \geq 3d(p-1) > 0.$$

Proof. The generic fibre is indecomposable and nonhyperelliptic, since these open conditions hold at Y . Over an algebraic closure, an indecomposable principally polarized abelian threefold is a Jacobian [OU73, Theorem 4]. A curve and an isomorphism of principally polarized abelian varieties therefore exist over a finite extension of $k(\mathbf{P}^1)$. Take the normalization of $\mathbf{P}^1$ in the finite extension, and make a further finite base change for stable reduction. After a further finite extension, choose a line bundle ξ of degree one whose fourth power is the canonical bundle of the generic curve; use it to define the Abel embedding. Since the Jacobian has good reduction everywhere, all stable fibres have compact type [OU73, Lemma 6]. This proves (i); it also gives $\delta_0 = 0$ and $\deg \lambda = d(p-1)$.

For (ii), Lemma 9.2 gives $\delta_1 \leq 3 \deg \lambda$. To compute the height, pass to the regular semistable resolution of $\mathcal{C}$. A node with local equation $xy = t^m$ contributes an edge of length m ; resolution replaces the edge by a rational chain and preserves both $\deg \lambda$ and the relative dualizing self-intersection. Thus δ_1 counts separating nodes with their thicknesses, including the contributions from ramified base change. For the dual graph of a curve of compact type, a separating edge of length ℓ , with genera i and $g-i$ on its two sides, contributes

$$\epsilon = \left(\frac{4i(g-i)}{g} - 1 \right) \ell, \quad \varphi = \frac{2i(g-i)}{g} \ell.$$

Substituting the formulas for ϵ and φ , together with $\omega_g^2 = 12 \deg \lambda - \delta$, in the canonical Gross–Schoen height formula [Zha10, Theorem 1.3.1, p. 5; Proposition 4.4.1 and its proof, pp. 58–59] gives, for $g = 3$,

$$\langle \Delta_\xi, \Delta_\xi \rangle = 21 \deg \lambda - 6\delta_1 \geq 3 \deg \lambda = 3d(p-1) > 0.$$

This function-field height calculation is valid over every constant field [Zha10, Section 2.1, p. 16]. $\square$

9.2. A correspondence detected by its height.

Lemma 9.4 (A primitive correspondence from a positive height). *Let k be an algebraically closed field of characteristic $p > 0$, let T/k be a smooth projective connected curve, and put $K = k(T)$. Let $\mathcal{C} \rightarrow T$ be a stable family of curves of genus three and compact type with smooth generic fibre, and let $\mathcal{J} \rightarrow T$ be its Jacobian abelian scheme with principal polarization θ . Let Y/k be an abelian threefold with polarization class η , and let $a : Y \times T \rightarrow \mathcal{J}$ be an isogeny. Use symmetric representatives for θ_K and η , and assume $a_K^* \theta_K = b\eta_K$ in $\mathrm{CH}^1(Y_K)_{\mathbf{Q}}$ for some $b \in \mathbf{Q}_{>0}$. Choose a line bundle of degree one ξ on $\mathcal{C}_K$ with $4\xi = K_{\mathcal{C}_K}$ in $\mathrm{Pic}(\mathcal{C}_K)_{\mathbf{Q}}$. If $h_{\mathrm{GS}} = \langle \Delta_{\xi}, \Delta_{\xi} \rangle > 0$, there is a correspondence $\Gamma \in \mathrm{CH}^2(T \times Y)_{\mathbf{Q}}$ satisfying the following.*

(i) *For every $\ell \neq p$,*

$$\mathrm{cl}_{\ell}(\Gamma) \in H^1(T, \mathbf{Q}_{\ell}) \otimes \ker(\eta \cup - : H^3(Y, \mathbf{Q}_{\ell}) \rightarrow H^5(Y, \mathbf{Q}_{\ell}(1))(2)).$$

(ii) *The self-intersection is*

$$\deg \Gamma^2 = \frac{\deg(a)}{6} h_{\mathrm{GS}} > 0. \quad (9.1)$$

Proof. We extend a homologically trivial cycle on the generic Jacobian, pull it back to $Y \times T$, and project to $H^1(T) \otimes H^3(Y)$. Write C_1 for the Beauville component of degree one of the Abel curve in $\mathcal{J}_K$, and set

$$z = \mathcal{F}_{\theta}(C_1) \in \mathrm{CH}^2(\mathcal{J}_K)_{\mathbf{Q}}.$$

Zhang's theorem [Zha10, Theorem 1.5.6, p. 14, and its proof, p. 71] gives

$$\theta_K z = 0, \quad \mathrm{cl}_{\ell}(z) = 0, \quad \langle z, z \rangle = h_{\mathrm{GS}}/6.$$

Put $z_Y = a_K^* z$. The identity $a_K^* \theta_K = b\eta_K$ gives $\eta_K z_Y = 0$ and $\mathrm{cl}_{\ell}(z_Y) = 0$. Choose a Chow extension $Z_{\mathcal{J}} \in \mathrm{CH}^2(\mathcal{J})_{\mathbf{Q}}$ of z . For $g : \mathcal{J} \rightarrow T$, its fibre classes define a section of the lisse sheaf $R^4 g_* \mathbf{Q}_{\ell}(2)$. This section vanishes at the geometric generic point and hence on every fibre. Each fibre restriction is therefore numerically trivial, so $Z_{\mathcal{J}}$ is an admissible extension in the definition of heights over function fields [Zha10, Section 2.1, pp. 16–17].

The height is the intersection number on the total space:

$$\langle z, z \rangle = \deg Z_{\mathcal{J}}^2. \quad (9.2)$$

To verify independence of the extension, let $Z'_{\mathcal{J}}$ be another Chow extension of z . Localization gives

$$Z'_{\mathcal{J}} - Z_{\mathcal{J}} = \sum_t (i_t)_* D_t, \quad D_t \in \mathrm{CH}^1(\mathcal{J}_t)_{\mathbf{Q}},$$

where $i_t : \mathcal{J}_t \hookrightarrow \mathcal{J}$ are fibre inclusions. The fibre restrictions of both extensions are numerically trivial. The projection formula therefore gives

$$\deg((Z'_{\mathcal{J}})^2 - Z_{\mathcal{J}}^2) = \sum_t \deg(D_t \cdot i_t^*(Z'_{\mathcal{J}} + Z_{\mathcal{J}})) = 0.$$

Thus the total intersection is independent of the admissible Chow extension.

The isogeny a is finite locally free. Set $Z = a^* Z_{\mathcal{J}}$. Its fibre restrictions are homologically and hence numerically trivial, so Z is admissible. The Chow projection formula and (9.2) give

$$\deg Z^2 = \deg(a) \deg Z_{\mathcal{J}}^2 = \frac{\deg(a)}{6} h_{\mathrm{GS}} > 0.$$

The vanishing of the fibre class removes the $H^0(T) \otimes H^4(Y)$ Künneth component of $\mathrm{cl}_{\ell}(Z)$. Write the remaining components as

$$\mathrm{cl}_{\ell}(Z) = c + v, \quad c \in H^1(T) \otimes H^3(Y)(2), \quad v \in H^2(T) \otimes H^2(Y)(2).$$

Both cv and v^2 vanish for degree reasons on T . Apply the algebraic Chow–Künneth projector π_Y^3 to the Y factor of Z and denote the result by Γ . Its class is c , so $\deg \Gamma^2 = \deg Z^2 > 0$, proving (ii).

Finally, $\eta_K z_Y = 0$ and localization show that $\mathrm{pr}_Y^* \eta \cdot Z$ is represented by vertical cycles. The $H^1(T) \otimes H^5(Y)$ component of a vertical class is zero. Hence $(1 \otimes \eta)c = 0$, proving (i). The positive self-intersection makes the cylinder map $H^1(T, \mathbf{Q}_\ell) \rightarrow H^3(Y, \mathbf{Q}_\ell)(1)$ nonzero for every $\ell \neq p$. All constructions use rational Chow correspondences. $\square$

9.3. Curves generating the third cohomology. We first generate the primitive cohomology at the geometric generic point, then specialize to every prescribed isogeny class. The curves and correspondences are chosen simultaneously for all $\ell \neq p$.

Proof of Theorem 1.2. Generation at the geometric generic point. Work in the moduli space $\mathcal{A}_{3,1,n}$ over $\overline{\mathbf{F}}_p$, with $n \geq 3$ prime to p , and let W be the reduced Newton stratum with slopes $0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1$. Every irreducible component has dimension four and generic a -number one [CO24, Section 2.5.1.6.4(a), p. 110; Section 2.9.5, pp. 155–156]. The decomposable locus of p -rank one is contained in the images of $\mathcal{A}_{1,f=1} \times \mathcal{A}_{2,f=0}$ and $\mathcal{A}_{1,f=0} \times \mathcal{A}_{2,f=1}$. Their dimensions are at most $1+1$ and $0+2$, respectively, by [CO24, Section 2.5.2.1, p. 113]. The locus of hyperelliptic Jacobians of genus three and p -rank one has dimension three: for odd p , use [GP05, Theorem 1 and Proposition 2] with its correction [GP10, Proposition 1]; for $p = 2$, use [PZ12, Corollary 1.3, p. 708]. Thus the geometric generic point of each component of W is indecomposable and nonhyperelliptic.

The stratum W is irreducible [CO11, Theorem A, p. 1359]. Its smooth locus $S = W^{\mathrm{sm}}$ is stable under prime-to- p Hecke correspondences: their two projections are étale and identify the corresponding strict local completions. The connected geometric ℓ -adic monodromy group of the universal abelian scheme over S is therefore $\mathrm{Sp}_{6, \mathbf{Q}_\ell}$ for every $\ell \neq p$ [Cha05, Proposition 4.5.4, p. 299].

Put $K = \overline{\mathbf{F}}_p(S)$ and fix an algebraic closure $\overline{K}$. Apply Lemmas 9.1 and 9.4 and Proposition 9.3 over $\overline{K}$ to the geometric generic fibre $Y_{\overline{\eta}}$, with principal polarization θ . The curve and correspondence descend to a finite extension L/K ; write them as T_L and Γ_L . Their image in

$$P_\ell = \ker(\theta : H^3(Y_{\overline{\eta}}, \mathbf{Q}_\ell) \longrightarrow H^5(Y_{\overline{\eta}}, \mathbf{Q}_\ell)(1))$$

is nonzero for every $\ell \neq p$.

Put $G_K = \mathrm{Aut}(\overline{K}/K)$ and $G_L = \mathrm{Aut}(\overline{K}/L)$. Since Γ_L is defined over L , its realization

$$(\Gamma_L)_* : H^1((T_L)_{\overline{K}}, \mathbf{Q}_\ell)(-1) \longrightarrow P_\ell$$

is G_L -equivariant. Its image I_ℓ is therefore G_L -stable. Let G_ℓ and $G_{\ell,L}$ be the Zariski closures of the images of G_K and G_L on $H^1(Y_{\overline{\eta}}, \mathbf{Q}_\ell)$. Since S is normal, G_K surjects onto $\pi_1(S, \overline{\eta})$. Thus $G_\ell^\circ = \mathrm{Sp}_{6, \mathbf{Q}_\ell}$. The subgroup G_L has finite index in G_K ; the purely inseparable part of L/K preserves the absolute Galois group. Consequently,

$$G_{\ell,L}^\circ = G_\ell^\circ = \mathrm{Sp}_6.$$

Indeed, finitely many translates of $G_{\ell,L}$ cover G_ℓ , so the two groups have the same identity component. The stabilizer of I_ℓ in $G_{\ell,L}$ is Zariski closed and contains the image of G_L . Thus I_ℓ is stable under Sp_6 . The representation P_ℓ is the irreducible primitive summand of dimension fourteen in the third exterior power of the standard representation. Since $I_\ell \neq 0$, we obtain $I_\ell = P_\ell$ for every $\ell \neq p$.

Simultaneous generation on one open subset. Choose an integral finite dominant model $S_L \rightarrow U$ of L/K , where $U \subset S$ is nonempty and open. Spread the curve, correspondence, and polarizations over S_L . Shrinking U removes the images of the bad loci under this finite map. We may thus assume that S_L is smooth over $\overline{\mathbf{F}}_p$, the curve is smooth and projective over S_L , and the cycle closures are flat of relative codimension two. The model, family, and cycles are independent of ℓ . For each $\ell \neq p$, the induced map of lisse sheaves has generic rank fourteen and hence is surjective onto the primitive part at every geometric point.

For $u \in U(\overline{\mathbf{F}}_p)$, choose $v \in S_L(\overline{\mathbf{F}}_p)$ above u and write T_v, Γ_v for the specialized curve and correspondence. Choose a smooth ample complete intersection curve C in the dual of Y_u . The Poincaré divisor on $C \times Y_u$, multiplied by the polarization of Y_u , gives a correspondence of codimension two with image $\theta H^1(Y_u)$. Weak Lefschetz and Poincaré duality prove the surjectivity onto $\theta H^1(Y_u)$. Together with Γ_v , this gives a correspondence from $T_v \sqcup C$ generating all of $H^3(Y_u)$.

Transfer to the prescribed isogeny class. The full polarized isogeny class of each point of $W(\overline{\mathbf{F}}_p)$ meets every central leaf in W [CO11, Proposition 4.2, p. 1375]. The prime-to- p Hecke orbit of a point is dense in its central leaf [CO24, Theorem 9.1.1, p. 551]. The full polarized isogeny class therefore contains a dense subset of each central leaf and is dense in W . Choose a principally polarized threefold Y_0 isogenous to the given Y [vdGM11, Chapter XI, Corollary 11.26, p. 170] and add level structure. Choose Y_u in the intersection of U with the full polarized isogeny class of Y_0 , together with the correspondence constructed at u . Choose an actual isogeny $a : Y \rightarrow Y_u$, clearing the denominator of a quasi-isogeny if necessary. Pullback of the constructed correspondence by $1 \times a$ gives the required Γ on $T \times Y$. The curves, cycles, and isogeny have finite presentation over $\overline{\mathbf{F}}_p$, so they descend together to a finite field. $\square$

10. THE MIXED DIHEDRAL CLASS

Let (S, T) be a dihedral pair as in Lemma 8.1, with S almost ordinary and T ordinary. After a finite extension, assume that all endomorphisms are defined, the normalized Frobenius group is torsion-free, and $q = Q^2$. Label the roots of S as

$$u_0, u_1 = \bar{u}_0, v_0, v_1 = \bar{v}_0,$$

with slopes $0, 1, \frac{1}{2}, \frac{1}{2}$. Put

$$z_{ij} = u_i v_j \quad (i, j \in \{0, 1\}).$$

In the notation $G = \langle r, s \rangle$ of Lemma 8.1, choose $H_S = \langle s \rangle$ and label

$$u_0, u_1, v_0, v_1 \quad \text{by} \quad H_S, r^2 H_S, r H_S, r^3 H_S,$$

respectively. The unordered adjacent pair $\{H_S, r H_S\}$ is stabilized by $\langle rs \rangle$, and its three conjugates give the other product labels. The stabilizer of the product character is therefore a reflection in the class opposite to H_S . Since S is neat, its normalized generators u_0/Q and v_0/Q are multiplicatively independent. Equality between two positive powers of distinct z_{ij} would give a nonzero relation between these generators. Hence the four product values remain distinct after every positive power. Put

$$K = \mathbf{Q}(z_{00}).$$

Then K is the quartic field fixed by the opposite reflection, and its four embeddings send z_{00} to the four values z_{ij} . Subtracting the slope of Q from the sums of the two labelled slopes gives

$$\left\{ \frac{v(z_{ij}/Q)}{v(q)} : i, j \in \{0, 1\} \right\} = \{0, 0, 1, 1\}.$$

The values z_{ij}/Q are therefore ordinary q -Weil integers. For the fixed normal closure and decomposition group, Lemma 8.1(v) gives one Weil germ in the ordinary reflection class. After a further constant extension, the roots of T may therefore be labelled

$$b_{ij} = z_{ij}/Q. \tag{10.1}$$

Indeed, the corresponding roots have the same principal divisors and absolute values, so their quotient is a root of unity by Kronecker's theorem. The extension makes these quotients equal to one. The four roots in (10.1) remain distinct after every further constant extension. Choose a supersingular elliptic curve E with Frobenius Q . All subsequent extensions preserve these labels.

Lemma 10.1 (The transcendental projector). *Let B/k be an abelian surface whose geometric endomorphism algebra is a quartic CM field, and assume that all geometric divisor classes are defined over k . Then $h^2(B)$ has a self-adjoint rational Chow projector whose realization has rank four and projects onto the orthogonal complement of $\mathrm{NS}(B)_{\mathbf{Q}}$.*

Proof. The polarization identifies $\mathrm{NS}(B)_{\mathbf{Q}}$ with the subspace of $\mathrm{End}^0(B)$ fixed by Rosati, which is the real quadratic subfield. Choose symmetric divisors D_1, D_2 forming a basis and let $M = (D_i \cdot D_j)_{i,j}$. The Hodge index theorem makes M invertible. For the Chow–Künneth projector π_2^B of [Anc21, Theorem 4.1(1), p. 14], put

$$p_{\mathrm{NS},B} = \sum_{i,j} (M^{-1})_{ij} \pi_2^B \circ [D_i \times D_j] \circ \pi_2^B, \quad p_{\mathrm{tr},B} = \pi_2^B - p_{\mathrm{NS},B}. \quad (10.2)$$

The first correspondence sends x to $\sum_{i,j} (x \cdot D_i)(M^{-1})_{ij} D_j$. Matrix multiplication gives idempotence in the Chow ring, and symmetry of M^{-1} gives self-adjointness. Thus $p_{\mathrm{tr},B}$ is the required complementary projector. Write $T_{\mathrm{tr}}^2(B) = (h^2(B), p_{\mathrm{tr},B})$. $\square$

Fix a good prime $\ell_0 \in s(S \times T)$ and a splitting field $\Omega/\mathbf{Q}_{\ell_0}$. For a motive N , write $N_{\ell_0} = R_{\ell_0}(N)$ and $N_{\Omega} = N_{\ell_0} \otimes \Omega$. Enlarge $k = \mathbf{F}_q$ so that the polarizations of S, T, E and the projectors of Lemma 10.1 are defined over k , retaining $q = Q^2$ and the curve E . Let F_q denote geometric Frobenius. Put

$$H = h^1(S), \quad W = h^1(T), \quad U = h^1(E), \quad V = T_{\mathrm{tr}}^2(S).$$

The two divisor lines of $H^2(S)_{\Omega}$ have root labels $u_0 \wedge u_1$ and $v_0 \wedge v_1$; their Frobenius eigenvalue is q . The remaining four lines form V_{Ω} , with eigenvalues z_{ij} . Put

$$M = V \otimes U(1), \quad R = V \otimes W \otimes U(2).$$

The projector of Lemma 10.1 makes M a Chow summand of $h^3(S \times E)(1)$. A polarization of T identifies $W^{\vee}$ with $W(1)$, hence $R \simeq M \otimes W^{\vee}$.

To algebraize the Frobenius-equivariant maps $W_{\ell_0} \rightarrow M_{\ell_0}$, we realize M_{ℓ_0} as a quotient of the cohomology of a curve. Frobenius semisimplicity lifts these maps to the curve cohomology, where Tate’s homomorphism theorem expresses them as $\mathbf{Q}_{\ell_0}$ -linear combinations of realizations of Chow morphisms.

Proposition 10.2. *The cycle class map*

$$\mathrm{Hom}_{\mathrm{CHM}(k)_{\mathbf{Q}}}(\mathbf{1}, R) \otimes_{\mathbf{Q}} \mathbf{Q}_{\ell_0} \longrightarrow R_{\ell_0}^{F_q=1}$$

is surjective after a finite extension of k .

Proof. Apply Theorem 1.2 to $Y = S \times E$. Project the resulting curve correspondence onto M . After extending k so that each curve component has a rational base point, its h^1 motive identifies with that of its Jacobian. We obtain an abelian variety J/k and a Chow morphism

$$s : h^1(J) \longrightarrow M$$

whose ℓ_0 -adic realization is surjective. Frobenius acts semisimply on $H^1(J, \mathbf{Q}_{\ell_0})$, so every Frobenius-equivariant map $t : W_{\ell_0} \rightarrow M_{\ell_0}$ lifts to a Frobenius-equivariant map $\tilde{t} : W_{\ell_0} \rightarrow H^1(J, \mathbf{Q}_{\ell_0})$. Tate’s homomorphism theorem [Tat66, Main Theorem, Section 1, p. 134] expresses $\tilde{t}$ as a $\mathbf{Q}_{\ell_0}$ -linear combination of pullbacks along homomorphisms $J \rightarrow T$. These pullbacks define Chow morphisms $W \rightarrow h^1(J)$. Composition with s therefore realizes t by algebraic correspondences. The identification $R \simeq M \otimes W^{\vee}$ completes the proof. $\square$

Use the symmetric square and the notation $a \odot b$ of (2.2).

Lemma 10.3. *There is a Chow morphism*

$$\mathcal{B} : R \otimes R \longrightarrow \mathrm{Sym}_{\mathrm{gr}}^2 H \otimes \mathrm{Sym}_{\mathrm{gr}}^2 W(2)$$

whose realization is

$$\mathcal{B}((v \otimes w \otimes e) \otimes (v' \otimes w' \otimes f)) = \psi_E(e, f) C_S(v, v') \otimes (w \odot w'), \quad (10.3)$$

where ψ_S and ψ_E are polarization pairings for S and E , and

$$\begin{aligned} C_S(a \wedge b, c \wedge d) &= \psi_S(b, d) a \odot c - \psi_S(b, c) a \odot d \\ &\quad - \psi_S(a, d) b \odot c + \psi_S(a, c) b \odot d. \end{aligned} \quad (10.4)$$

Proof. The polarization ψ_S , tensor symmetries, and the exterior-square projectors define $C_S : V \otimes V \rightarrow \mathrm{Sym}_{\mathrm{gr}}^2 H \otimes \mathbf{L}$ with realization (10.4). Regroup the factors of $(VWU)(VWU)$ as $(VV)(WW)(UU)$ and apply C_S , the projector (2.2) on $W^{\otimes 2}$, and ψ_E . The regrouping crosses the odd factors U and W once. Multiplying this composite by -1 gives (10.3). The two polarization factors contribute $\mathbf{L}^{\otimes 2}$, and $\mathbf{L}^{\otimes 2}(4) = \mathbf{Q}(2)$ gives the stated target. $\square$

Proposition 10.4. *There is a cycle $\xi \in \mathrm{CH}^2((S \times T)^2)_{\mathbf{Q}}$ whose ℓ_0 -adic class over Ω has exactly four nonzero weight components, with normalized characters*

$$\pm r_1, \quad \pm r_2, \quad r_1 = (2, 0, -1, -1), \quad r_2 = (0, 2, -1, 1). \quad (10.5)$$

Proof. Nonzero contractions. Choose nonzero eigenvectors $\mathbf{u}_i, \mathbf{v}_j$ of H_{Ω} and $\mathbf{b}_{ij}$ of W_{Ω} with the root labels of (10.1). Set

$$a_{ij} = \mathbf{u}_i \wedge \mathbf{v}_j, \quad w_{ij} = \mathbf{b}_{1-i, 1-j}, \quad t_{ij}(e) = a_{ij} \otimes w_{ij} \otimes e.$$

Tate twists are suppressed in these vector formulas. The eigenvalue of w_{ij} is q/b_{ij} , so $u_i v_j (q/b_{ij}) Q = q^2$. The four products $u_i v_j$ are distinct, and consequently

$$R_{\Omega}^{F_q=1} = \bigoplus_{i,j \in \{0,1\}} \Omega(a_{ij} \otimes w_{ij}) \otimes U_{\Omega}. \quad (10.6)$$

Choose $e, f \in U_{\Omega}$ with $\psi_E(e, f) \neq 0$. The perfection and Frobenius compatibility of ψ_S give $\psi_S(\mathbf{u}_0, \mathbf{u}_1) \neq 0$ and $\psi_S(\mathbf{v}_0, \mathbf{v}_1) \neq 0$. Since $u_i v_j \neq q$, Frobenius compatibility gives $\psi_S(\mathbf{u}_i, \mathbf{v}_j) = 0$; alternation gives $\psi_S(\mathbf{u}_i, \mathbf{u}_i) = 0$. Thus (10.4) gives, for example,

$$C_S(\mathbf{u}_i \wedge \mathbf{v}_0, \mathbf{u}_i \wedge \mathbf{v}_1) = \psi_S(\mathbf{v}_0, \mathbf{v}_1) \mathbf{u}_i \odot \mathbf{u}_i.$$

This gives the repeated weight of $\mathbf{u}_i$. Equations (10.3) and (10.4) give nonzero multiples of

$$\mathbf{u}_i \odot \mathbf{u}_i \otimes (w_{i0} \odot w_{i1}), \quad \mathbf{v}_j \odot \mathbf{v}_j \otimes (w_{0j} \odot w_{1j}). \quad (10.7)$$

Set

$$x_1 = u_0/Q, \quad x_2 = v_0/Q, \quad y_1 = b_{00}/Q, \quad y_2 = b_{01}/Q.$$

Then (10.1) gives

$$y_1 = x_1 x_2, \quad y_2 = x_1 x_2^{-1}. \quad (10.8)$$

With $\beta_i = x_i^2$ and $\gamma_i = y_i^2$, (10.8) gives

$$\gamma_1 = \beta_1 \beta_2, \quad \gamma_2 = \beta_1 \beta_2^{-1}. \quad (10.9)$$

The four tensors in (10.7) have weights $r_1, -r_1, r_2, -r_2$, respectively.

Choice of a rational cycle. By Proposition 10.2, choose rational cycles $Z_1, \dots, Z_N : \mathbf{1} \rightarrow R$ whose classes span $R_{\ell_0}^{F_q=1}$. Bilinearity and the nonzero contractions in (10.7) show that the rational span of the cycles $\mathcal{B}(Z_a \otimes Z_b)$ has nonzero projection onto each of the four weight spaces. The four projection kernels are proper rational linear subspaces of this finite-dimensional rational span. Since $\mathbf{Q}$ is infinite, their complement contains a rational cycle Ξ .

The remaining weights. Formula (10.4) vanishes on (a_{ij}, a_{ij}) . Pairs with exactly one index in common give the four weights in (10.5); pairs $(a_{ij}, a_{1-i,1-j})$ give weight zero. Thus Ξ has support in $\{0, \pm r_1, \pm r_2\}$.

Removal of the zero weight. Let f_S act by q -Frobenius on the two S factors of $(S \times T)^2$ and by the identity on the two T factors. Its eigenvalue on the zero-weight component is q , and its eigenvalues on the other four components are $u_0^2, u_1^2, v_0^2, v_1^2$. These eigenvalues come from the S factors. The total Frobenius acts as one on the Tate classes of the twisted tensor. Each of $u_0^2, u_1^2, v_0^2, v_1^2$ differs from q by the multiplicative independence of x_1, x_2 . The cycle

$$\xi = (f_S^* - q \text{id})\Xi$$

has support exactly $\{\pm r_1, \pm r_2\}$. Its codimension is two because the target of $\mathcal{B}$ is a Chow summand of $h^4((S \times T)^2)(2)$. $\square$

Proposition 10.5 (Integral generation by the constructed weights). *For $A = S \times T$, the weights r_1, r_2 in Proposition 10.4 form a $\mathbf{Z}$ -basis of the even character kernel:*

$$R_{\text{ev}} = \mathbf{Z}r_1 \oplus \mathbf{Z}r_2. \quad (10.10)$$

Proof. In the coordinates of (10.8), put

$$\Lambda_{\text{raw}} = \ker(\mathbf{Z}^4 \longrightarrow \langle x_1, x_2, y_1, y_2 \rangle, (m_1, m_2, n_1, n_2) \longmapsto x_1^{m_1} x_2^{m_2} y_1^{n_1} y_2^{n_2}).$$

The group of normalized Frobenius values is torsion-free over the chosen constant extension. Thus Lemma 5.2(ii) identifies Λ_{raw} with Λ_A^{red} and gives $R_{\text{ev}} = \Lambda_{\text{raw}} \cap X^*(\bar{L}')$. The neatness of S makes x_1, x_2 multiplicatively independent, so (10.8) yields

$$\Lambda_{\text{raw}} = \mathbf{Z}s_1 \oplus \mathbf{Z}s_2, \quad s_1 = (1, 1, -1, 0), \quad s_2 = (1, -1, 0, -1).$$

Both generators have odd parity. Since $r_1 = s_1 + s_2$ and $r_2 = s_1 - s_2$, the even kernel is

$$R_{\text{ev}} = \{as_1 + bs_2 : a \equiv b \pmod{2}\} = \mathbf{Z}r_1 \oplus \mathbf{Z}r_2.$$

Indeed, the coefficients of $as_1 + bs_2$ in this basis are $(a+b)/2$ and $(a-b)/2$. Evaluation on normalized Frobenius maps the even character lattice to a torsion-free group. Its kernel (10.10) is therefore saturated in $X^*(\bar{L}')$. The index $[\Lambda_{\text{raw}} : R_{\text{ev}}] = 2$ records the passage to even characters; the ambient lattice for saturation is $X^*(\bar{L}')$. $\square$

Theorem 10.6. *For every $n \geq 1$ and every $\ell \neq p$, every Tate class on $(S \times T)^n$ is algebraic.*

Proof. By Proposition 10.4, a rational cycle has a nonzero component in each of $\pm r_1, \pm r_2$. These weights generate the even character kernel over $\mathbf{Z}$ by Proposition 10.5. Apply Corollary 5.5 at ℓ_0 . Through Proposition 5.4, it gives the Tate and folklore conjectures for every power and every $\ell \neq p$. The constructions descend to a finite extension of the constant field, and Lemma 2.1 descends invariant classes. $\square$

Proof of Theorem 5.1. Suppose first that $\dim A = 4$. Geometrically simple fourfolds satisfy the assertion by Theorem 6.2. The partitions $3 + 1$, $2 + 1 + 1$, and $1 + 1 + 1 + 1$ are treated in Sections 7.1 and 7.2. For two surface factors, Section 7.3 treats $d(S, T) = 0, 1$, repeated factors, and supersingular factors. A pair satisfying (7.3) is dihedral by Lemma 8.1, and Theorem 10.6 proves the assertion for all its powers. These possibilities exhaust the geometric isogeny types of fourfolds. Isogenies preserve rational Chow motives and the Tate conjecture.

For $\dim A = 0$, every power is a point. If $0 < d = \dim A < 4$, choose an elliptic curve $E/\bar{\mathbf{F}}_p$ and put $B = A \times E^{4-d}$. This fourfold has an elliptic isogeny factor, so Sections 7.1 and 7.2 apply. Projection and insertion of the origin of E^{4-d} exhibit $h(A^n)$ as an algebraic direct summand of $h(B^n)$ for every n . Pull back a Tate class on A^n to B^n , represent it by a cycle, and restrict that cycle along the inclusion. This proves the assertion for A^n . $\square$

Proof of Theorem 1.1. Apply Theorem 5.1 over $\overline{\mathbf{F}}_q$. The finitely many cycles representing an invariant class are defined over a finite extension. By Lemma 2.1, their $\mathbf{Q}_\ell$ -linear combination descends to $\mathbf{F}_q$. $\square$

11. APPLICATIONS

11.1. Consequences of the theorem on all powers.

Proposition 11.1. *Let $A/\mathbf{F}_q$ be an abelian variety of dimension at most four, and let M be a rational Chow direct summand of*

$$\bigoplus_{j=1}^u h(A^{m_j})(a_j), \quad m_j \geq 0, \quad a_j \in \mathbf{Z},$$

with summand maps defined over $\mathbf{F}_q$. For $r \in \mathbf{Z}$ and $\ell \neq p$, consider the realization map

$$\mathrm{cl}_{\ell, M} : \mathrm{Hom}_{\mathrm{CHM}(\mathbf{F}_q)\mathbf{Q}}(\mathbf{1}, M(r)) \otimes_{\mathbf{Q}} \mathbf{Q}_\ell \longrightarrow R_\ell(M(r))^{F_q=1}.$$

- (i) *The map $\mathrm{cl}_{\ell, M}$ is surjective.*
- (ii) *Its kernel is*

$$\ker \mathrm{cl}_{\ell, M} = \mathcal{N}(\mathbf{1}, M(r)) \otimes_{\mathbf{Q}} \mathbf{Q}_\ell,$$

where $\mathcal{N}$ is the ideal of numerically trivial morphisms.

Proof. Write $i : M \rightarrow N$ and $e : N \rightarrow M$ for the summand maps, with $ei = 1_M$. For $t_j = a_j + r$, purity gives

$$R_\ell(h(A^{m_j})(t_j))^{F_q=1} = H_{\mathrm{ét}}^{2t_j}(A_{\overline{\mathbf{F}}_q}^{m_j}, \mathbf{Q}_\ell(t_j))^{F_q=1}.$$

This space and $\mathrm{CH}^{t_j}(A^{m_j})_{\mathbf{Q}}$ vanish outside the possible codimensions. In the remaining codimensions, Theorem 1.1 gives an algebraic representative for each component of an invariant class. Applying e proves (i).

For (ii), semisimple Frobenius and Poincaré duality give a perfect pairing between invariant classes in complementary codimensions on every power of A . Theorem 1.1 makes both spaces algebraic, so a numerically trivial cycle has zero realization. If $z : \mathbf{1} \rightarrow M(r)$ is numerically trivial, then iz is numerically trivial, since $\mathcal{N}$ is a two-sided ideal. Thus $R_\ell(iz) = 0$, and applying $R_\ell(e)$ gives $R_\ell(z) = 0$.

Conversely, the composition pairing between the numerical quotients of $\mathrm{Hom}(\mathbf{1}, M(r))$ and $\mathrm{Hom}(M(r), \mathbf{1})$ is nondegenerate and remains so after extension to $\mathbf{Q}_\ell$. An element with zero realization pairs to zero with every complementary morphism. It therefore belongs to $\mathcal{N}(\mathbf{1}, M(r)) \otimes_{\mathbf{Q}} \mathbf{Q}_\ell$. $\square$

Proposition 11.1 applies to every object of the pseudoabelian rigid tensor category generated by $h(A)$ and the Tate motive, and to every smooth projective variety whose Chow motive belongs to this category.

Corollary 11.2. *Let $A_1, \dots, A_s/\mathbf{F}_q$ be abelian varieties with $\sum_i \dim A_i \leq 4$, and put $Y = \prod_i A_i^{m_i}$ for $m_i \geq 0$.*

- (i) *For every $e \geq 1$ and $\ell \neq p$, the variety $Y_{\mathbf{F}_{q^e}}$ satisfies the Tate conjecture in every codimension.*
- (ii) *Numerical and ℓ -adic homological equivalence agree on $\mathrm{CH}^r(Y_{\overline{\mathbf{F}}_q})_{\mathbf{Q}}$ for every r and $\ell \neq p$. The kernel of the cycle class map is therefore independent of ℓ .*
- (iii) *For every $e \geq 1$ and $0 \leq r \leq \dim Y$, the order of the pole of $\zeta(Y_{\mathbf{F}_{q^e}}, s)$ at $s = r$ is*

$$\dim_{\mathbf{Q}} N^r(Y_{\mathbf{F}_{q^e}}) = \dim_{\mathbf{Q}_\ell} H_{\text{ét}}^{2r}(Y_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(r))^{F_q^e=1},$$

where N^r denotes cycles over $\mathbf{F}_{q^e}$ modulo numerical equivalence.

Proof. Put $B = \prod_i A_i$ and $m = \max\{1, m_1, \dots, m_s\}$. Projection and insertion of origins make $h(Y)$ a rational Chow direct summand of $h(B^m)$, with $\dim B \leq 4$. Applying Proposition 11.1(i) over each finite extension proves (i).

For (ii), let $d = \dim Y$ and let $z \in \mathrm{CH}^r(Y_{\overline{\mathbf{F}}_q})_{\mathbf{Q}}$ be numerically trivial. The cycle z is defined over some $\mathbf{F}_{q^e}$. Semisimple Frobenius and Poincaré duality pair the F_q^e -fixed subspaces in codimensions r and $d - r$ perfectly. By (i), the complementary invariant classes are algebraic. Hence $\mathrm{cl}_\ell(z) = 0$. Homological triviality implies numerical triviality, proving (ii).

For (iii), the trace formula and the Weil bounds identify the pole order with the multiplicity of q^r as an eigenvalue of F_q^e on $H_{\text{ét}}^{2r}(Y_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell)$. Semisimplicity identifies this multiplicity with the dimension of the F_q^e -fixed subspace of $H_{\text{ét}}^{2r}(Y_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(r))$. By Proposition 11.1(ii), the cycle class map on $N^r(Y_{\mathbf{F}_{q^e}}) \otimes_{\mathbf{Q}} \mathbf{Q}_\ell$ is injective, and (i) makes it surjective. This proves (iii). $\square$

11.2. Consequences of Broe's theorem. Broe's theorem on standard conjecture D, stated in Corollary 4.2, gives the following consequences.

Smooth ample divisors.

Corollary 11.3. *Let A/k be an abelian fourfold over an algebraically closed field of characteristic $p > 0$, and let $i : T \hookrightarrow A$ be a smooth ample divisor. Numerical and ℓ -adic homological equivalence agree on $\mathrm{CH}^r(T)_{\mathbf{Q}}$ for every r and every $\ell \neq p$.*

Proof. Weak Lefschetz and Poincaré duality give

$$i^* : H^2(A, \mathbf{Q}_\ell(1)) \xrightarrow{\sim} H^2(T, \mathbf{Q}_\ell(1)), \quad (11.1)$$

$$i_* : H^4(T, \mathbf{Q}_\ell(2)) \xrightarrow{\sim} H^6(A, \mathbf{Q}_\ell(3)). \quad (11.2)$$

If $\alpha \in \mathrm{CH}^2(T)_{\mathbf{Q}}$ is numerically trivial, then for every divisor L on A ,

$$\deg_A(i_* \alpha \cdot L) = \deg_T(\alpha \cdot i^* L) = 0.$$

Thus $i_* \alpha$ is numerically trivial in $\mathrm{CH}^3(A)_{\mathbf{Q}}$, and its class vanishes by Corollary 4.2. The injectivity of (11.2) gives $\mathrm{cl}_\ell(\alpha) = 0$. For divisors, numerical triviality implies algebraic triviality up to torsion, and the Kummer sequence gives a zero class. Zero-cycles are controlled by degree. Homological triviality always implies numerical triviality. $\square$

Products of surfaces dominated by abelian surfaces.

Lemma 11.4. *Let $i : M \rightarrow N$ and $r : N \rightarrow M$ be maps of rational Chow motives with $ri = 1_M$. If D_ℓ holds for algebraic classes on N , it holds for algebraic classes on M .*

Proof. For a numerically trivial $z : \mathbf{1} \rightarrow M(q)$, the composition iz is numerically trivial, so its realization vanishes. Applying $R_\ell(r)$ gives $R_\ell(z) = 0$. $\square$

Theorem 11.5. *Let $S, T/k$ be smooth projective connected surfaces over an algebraically closed field of characteristic $p > 0$. Suppose abelian surfaces A, B admit dominant rational maps $A \dashrightarrow S$ and $B \dashrightarrow T$. Then $D_\ell(S \times T)$ holds in every codimension for every $\ell \neq p$.*

Proof. Successive blowups at points resolve these rational maps [Sta26, Lemma 54.4.3, Tag 0C5H], giving proper generically finite morphisms $f : A' \rightarrow S$ and $g : B' \rightarrow T$. For the total degree d_f , graph correspondences satisfy

$$\Gamma_f \circ {}^t\Gamma_f = d_f \Delta_S, \quad (11.3)$$

since $(f \times f)_*[\Delta_{A'}] = d_f[\Delta_S]$. Thus $h(S)$ is a rational Chow direct summand of $h(A')$, and $h(S \times T)$ is a summand of $h(A' \times B')$.

For the blowup $b : \tilde{X} \rightarrow X$ at a point of a smooth surface, the exceptional divisor E satisfies $E^2 = -1$ and $b_*E = 0$. The difference $\Delta_{\tilde{X}} - {}^t\Gamma_b \circ \Gamma_b$ is supported on $E \times E$ and equals $-E \times E$, the projector onto $\mathbf{L}$. Iterating gives

$$h(A') \simeq h(A) \oplus \mathbf{L}^{\oplus a}, \quad h(B') \simeq h(B) \oplus \mathbf{L}^{\oplus b}, \quad (11.4)$$

where a, b are the numbers of blowups. Consequently $h(A' \times B')$ is a direct sum of $h(A \times B)$, Tate twists of $h(A)$ and $h(B)$, and Tate motives. Each summand satisfies D_ℓ by Corollary 4.2. Apply Lemma 11.4. $\square$

Corollary 11.6. *Let A, B be abelian surfaces over an algebraically closed field of characteristic $p > 0$, and let S, T be smooth projective surfaces birational to $A/\{\pm 1\}$ and $B/\{\pm 1\}$, respectively. Then $D_\ell(S \times T)$ holds for every $\ell \neq p$.*

Proof. The quotient maps give dominant rational maps $A \dashrightarrow S$ and $B \dashrightarrow T$. Apply Theorem 11.5. $\square$

ACKNOWLEDGEMENTS

The author used OpenAI Codex for literature searches, exploration of proof strategies, reference verification, proof auditing, and LaTeX formatting.

APPENDIX A. DIHEDRAL DIVISOR MODULES

We compute the divisor modules for

$$D_4 = \langle r, s : r^4 = s^2 = 1, \, srs = r^{-1} \rangle, \quad c = r^2.$$

All permutation modules carry the left G -action $h[gH] = [hgH]$ on the left cosets G/H . For $D = 1$, a divisor $\sum_g \epsilon(g)[g]$ has coefficient action $(h\epsilon)(g) = \epsilon(h^{-1}g)$. Thus an H -fixed coefficient function is constant on the right cosets $H \backslash G$, identified with G/H by $Hg \mapsto g^{-1}H$.

First take $D = 1$. A normalized ordinary divisor is a sign function $\epsilon : G \rightarrow \{\pm 1\}$ satisfying $\epsilon(cg) = -\epsilon(g)$. If the quartic field is L^H , the function must also be H -fixed. For a fixed reflection H , its values on the four right cosets $H \backslash G$ are determined by two independent signs, because c pairs the four cosets. Hence there are exactly four functions for each of the four reflections, and sixteen in total. Each has stabilizer exactly its indicated reflection. A function is determined by its values on the ordered half $(1, s, r, rs)$ of G ; its values on (c, cs, cr, crs) are their negatives. In these coordinates the action is

$$\begin{aligned} r(x_1, x_2, x_3, x_4) &= (-x_3, -x_4, x_1, x_2), \\ s(x_1, x_2, x_3, x_4) &= (x_2, x_1, -x_4, -x_3). \end{aligned} \quad (A.1)$$

The table lists the sixteen functions. Each row consists of $\{\pm v, \pm w\}$, and $a \bmod 2$ specifies the conjugacy class of their two stabilizing reflections $\langle r^a s \rangle$.

P	$a \bmod 2$	v	w
1	0	$(-1, -1, -1, 1)$	$(-1, 1, 1, 1)$
2	0	$(-1, -1, 1, -1)$	$(-1, 1, -1, -1)$
3	1	$(-1, -1, -1, -1)$	$(-1, -1, 1, 1)$
4	1	$(-1, 1, -1, 1)$	$(-1, 1, 1, -1)$

(A.2)

The image of $d_{\pi, H}$ is the rational span of the G -translates of $\epsilon = \text{Div}_p(\pi^2/q)$. By (A.1), each set $\{\pm v, \pm w\}$ is one G -orbit. Row reduction gives its span:

$$\begin{aligned} P_1 &= \langle (1, 0, 0, -1), (0, 1, 1, 0) \rangle, & P_2 &= \langle (1, 0, 0, 1), (0, 1, -1, 0) \rangle, \\ P_3 &= \langle (1, 1, 0, 0), (0, 0, 1, 1) \rangle, & P_4 &= \langle (1, -1, 0, 0), (0, 0, 1, -1) \rangle. \end{aligned} \quad (\text{A.3})$$

The four planes in (A.3) are distinct and have dimension two. Hence two functions in (A.2) have the same divisor plane exactly when they are Galois conjugate.

For $D = \langle s \rangle$, put $f_j = r^j D$. Then $A = f_0 - f_2$ and $B = f_1 - f_3$ form a basis of $\mathbf{Q}[G/D]^-$.

If $H = \langle s \rangle$, then D acts on G/H by $j \mapsto -j$ for $r^j H$. Its orbits are $\{0\}$, $\{2\}$, and $\{1, 3\}$. The first two slopes are integral and complementary. The last orbit is preserved by c , so the Weil symmetry forces its slope to be $1/2$. Thus, up to complementation, the only nonconstant slope multiset is $(0, 1, \frac{1}{2}, \frac{1}{2})$, and its normalized divisor is A .

If $H = \langle rs \rangle$, then D acts by $j \mapsto 3 - j$. Its two orbits are $\{0, 3\}$ and $\{1, 2\}$. Local integrality allows the slope on the first orbit to be $0, \frac{1}{2}$, or 1 , and Weil symmetry gives the complementary slope on the second. The middle choice is the constant function with value $1/2$ and has zero normalized divisor. The two choices of rank two are complements, with ordinary multiset $(0, 0, 1, 1)$ and normalized divisors $\pm(A + B)$. Conjugating H treats the other reflection in each conjugacy class, so the two complementary choices in either class form one G -orbit.

For $H = \langle s \rangle$ or $H = \langle rs \rangle$, put $e_j = r^j H$, $a = e_0 - e_2$, and $b = e_1 - e_3$; then a, b are a basis of $\mathbf{Q}[G/H]^-$. Since $\tilde{d}(e_0)$ is respectively A and $A + B$, while $\tilde{d}(e_2) = -\tilde{d}(e_0)$ and $b = ra$, equivariance gives

H	$d(a)$	$d(b)$	$\det d$
$\langle s \rangle$	$2A$	$2B$	4
$\langle rs \rangle$	$2(A + B)$	$2(-A + B)$	8

Both maps have image $\mathbf{Q}[G/D]^-$, proving the converse in Lemma 8.1.

REFERENCES

- [Anc21] G. Ancona, *Standard conjectures for abelian fourfolds*, Invent. Math. **223** (2021), no. 1, 149–212, Numbering: <https://arxiv.org/abs/1806.03216v3>.
- [APFV25] S. Arango-Piñeros, S. Frengley, and S. Vemulapalli, *Galois groups of simple abelian varieties over finite fields and exceptional Tate classes*, 2025, Preprint, arXiv:2505.09589v1.
- [Bro26] Matt Broe, *The Tate conjecture for abelian fourfolds over finite fields*, 2026, Preprint, arXiv:2608.28651v1.
- [Cha05] Ching-Li Chai, *Monodromy of Hecke-invariant subvarieties*, Pure Appl. Math. Q. **1** (2005), no. 2, 291–303.
- [CO11] Ching-Li Chai and Frans Oort, *Monodromy and irreducibility of leaves*, Ann. of Math. (2) **173** (2011), no. 3, 1359–1396.
- [CO24] Ching-Li Chai and Frans Oort, *Hecke orbits*, 2024, Book manuscript, version of 25 February 2024.
- [FF26] S. Floccari and L. Fu, *The Hodge conjecture for Weil fourfolds with discriminant 1 via singular OG6-varieties*, J. Math. Pures Appl. (9) **210** (2026), 103876, Numbering: final author version, arXiv:2504.13607v2.
- [GP05] Darren Glass and Rachel Pries, *Hyperelliptic curves with prescribed p -torsion*, Manuscripta Math. **117** (2005), no. 3, 299–317, Theorem numbering follows the journal version; see also the 2010 erratum.
- [GP10] Darren Glass and Rachel Pries, *Erratum to: Hyperelliptic curves with prescribed p -torsion*, Manuscripta Math. **133** (2010), no. 3–4, 545–546.

- [Mar25] E. Markman, *Cycles on abelian $2n$ -folds of Weil type from secant sheaves on abelian n -folds*, 2025, Preprint, arXiv:2502.03415v2.
- [Mil22] J. S. Milne, *The Tate and standard conjectures for certain abelian varieties*, 2022, Preprint, arXiv:2112.12815v2. Numbering follows the PDF version of 6 February 2022.
- [MZ99] B. J. J. Moonen and Yu. G. Zarhin, *Hodge classes on abelian varieties of low dimension*, Math. Ann. **315** (1999), no. 4, 711–733, Numbering: <https://arxiv.org/abs/math/9901113v2>.
- [Orr15] Martin Orr, *Lower bounds for ranks of Mumford–Tate groups*, Bull. Soc. Math. France **143** (2015), no. 2, 229–246.
- [OU73] Frans Oort and Kenji Ueno, *Principally polarized abelian varieties of dimension two or three are Jacobian varieties*, J. Fac. Sci. Univ. Tokyo Sect. IA Math. **20** (1973), no. 3, 377–381, Numbering: Aarhus preprint, February 1973.
- [PZ12] Rachel Pries and Hui June Zhu, *The p -rank stratification of Artin–Schreier curves*, Ann. Inst. Fourier (Grenoble) **62** (2012), no. 2, 707–726.
- [RS22] Damian Rössler and Stefan Schröer, *Moret–Bailly families and non-liftable schemes*, Algebr. Geom. **9** (2022), no. 1, 93–121.
- [Sta26] Stacks Project Authors, *The Stacks Project*, 2026, Lemma 54.4.3, Tag 0C5H, accessed 10 September 2026.
- [Tat66] J. Tate, *Endomorphisms of abelian varieties over finite fields*, Invent. Math. **2** (1966), no. 2, 134–144.
- [vdGM11] Gerard van der Geer and Ben Moonen, *Abelian varieties*, 2011, Preliminary book manuscript, Chapter XI: Polarizations and Weil pairings, version of 15 September 2011.
- [Wat69] W. C. Waterhouse, *Abelian varieties over finite fields*, Ann. Sci. École Norm. Sup. (4) **2** (1969), no. 4, 521–560.
- [Zar15] Y. G. Zarhin, *Eigenvalues of Frobenius endomorphisms of abelian varieties of low dimension*, J. Pure Appl. Algebra **219** (2015), no. 6, 2076–2098, Numbering: <https://arxiv.org/abs/1312.0377v4>.
- [Zha10] Shou-Wu Zhang, *Gross–Schoen cycles and dualising sheaves*, Invent. Math. **179** (2010), no. 1, 1–73.